ANALYZING AND FORECASTING GLACIER MASS LOSS IN THE HIMALAYAS USING MACHINE LEARNING MODELS

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ABSTRACT

High Mountain Asia has emerged as a regional hotspot for glacier mass loss, which is a major issue for water supply, regional hydrology and glacier hazards. In this study, data analytics and visualization are used to examine glacier properties in High Mountain Asia and a basis for future forecasting is established. Randolph Glacier Inventory (RGI) v7.0 Regions 13, 14, and 15 were used to represent Central Asia, South Asia West and South Asia East, respectively. The data were cleaned, merged and analyzed in Power BI to analyze the count, area, elevation, and size class distribution of glaciers over the region. In addition, glacier mass-balance data were used to analyze mass-balance trends in the annual time-scale between 1990 and 2025 and to facilitate interpretation of future glacier mass loss with the aid of forecasts.

The results reveal distinct regional variations in glacier distribution, area, elevation and mass-balance dynamics in High Mountain Asia. The analysis reveals that spatial variability in glacier mass loss is not uniform, reflecting the differences found in glacier characteristics by region, by glacier size and by elevation. The results also indicate that areas of increased glacier-loss risk can be identified by negative mass-balance trends at an annual scale. In general, this study illustrates that Power BI visual analytics and time-series analysis can be useful tools to understand the change of glaciers, and to compare the glacier characteristics in the regions, and to support future glacier mass loss forecast based on climate changes.

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1. INTRODUCTION

1.1 Introduction

Glaciers are some of the most visible signs of climate change, and their accelerated melting is a cause for concern. In High Mountain Asia, glaciers are critical for regulating water availability by storing and releasing fresh water to supply water for downstream communities, agriculture, hydropower and ecosystems. But increasing temperature, uncertain precipitation patterns and shift in the timing of snowfall have guided to glacier mass loss acceleration across the region. These shifts have significant consequences for water security, glacier-fed rivers, and glacier related hazards, such as glacial lake outburst floods (GLOFs).

While previous research indicates that glacier mass loss is extensive across High Mountain Asia, the response across glaciers is highly variable. Rates of thinning and retreat vary between glaciers due to the variations in elevation, topography, debris cover, and regional-scale climate forcing. This variability makes glacier analysis more challenging and suggests a need for a data-driven regional approach that can discern regional-scale trends and variability.

The increasing availability of glacier inventories, satellite data, and climate reanalysis data offers valuable opportunities for data analytics in the investigation of glacier change. Data visualization and time series analysis can help to identify long-term trends and regional differences in mass loss, as well as potentially helpful forecasting patterns. As part of this approach, the current study engages in a data-driven analysis to explore glacier characteristics and patterns of glacier mass-loss in High Mountain Asia in support of better understanding of past changes and glacier projections. This project is an extension of the overall proposal goal to investigate trends in glacier melting using secondary data, visualization and forecasting.

1.2 Problem Statement

The altered environmental conditions and climate change have led to a faster glacier mass loss in High Mountain Asia in the recent decades. Glacier change in this region has major implications for water resources, hydrology, ecosystems, and the risk of hazards downstream The response of glacier in High Mountain Asia is not uniform in space as glacier characteristics differs according to the elevation, glacier size, regional climate, topography, and ablation process.

While the change in glaciers in particular regions of the Himalayas and High Mountain Asia has been studied extensively in the past, an integrated data-driven analysis that combines glacier inventory analysis, regional comparisons, visualization and analysis for forecasting is still needed. While large-scale glacier data sets like the Randolph Glacier Inventory (RGI) and glacier mass-balance data sets provide a wealth of environmental information, these data sets can be complex and difficult to analyze and visualize without effective techniques.

Additionally, most glacier research is conducted in the field of physical glaciology and climate science, with fewer studies using data analytics, machine learning and interactive visualization to look for glacier trends, to compare glacier behavior regionally, and to facilitate future glacier mass loss prediction. This dissertation thus seeks to fill this gap with the development of a structured analytical framework, which integrates glacier inventory information with mass-balance observations and glacier forecasting models to investigate glacier characteristics and patterns of glacier mass-loss in the High Mountain Asia domain.

1.3 Business Understanding

Although this study is more environmental than commercial, the concept of business understanding is applied to the real world problem, stakeholders, and practical value of the project. This step is most important in the data analytics field because it defines why the analysis matters, who can take benefit from it and what decisions the result can support.

The central issue in this dissertation is the growing trend of glacier mass loss in High Mountain Asia and its challenge to consistently interpret this change over a vast and climatically diverse mountain region. High Mountain Asia is mainly referred as a major freshwater reserve outside of the polar regions, which makes glacier change both scientifically and socio-economically important. Glacier mass changes can affect water resources, runoff variability, ecosystems, and risk downstream. Consequently, the analysis is relevant to environmental monitoring, climate change adaptation, water-resource management and hazard risk.

The primary stakeholders in this study includes climatologists, environmental managers, hydrologists, policy makers and regional bodies interested in glacier monitoring and water-resource management. From an analytics viewpoint, the project aims to provide insight from raw glacier inventory and other environmental data through descriptive, comparative and forecasting-focused analytics. The project's success will be measured in terms of providing a coherent picture of glacier trends in High Mountain Asia, and show how data analytics and visualization techniques can used to support the decision making for the environment.

1.3 Aims and Objectives

The main aim of this dissertation is to analyze the characteristics of glacier and glacier mass-loss patterns across High Mountain Asia using data-driven methods, and to establish a framework for forecasting.

The dissertation will have the following objectives:

- To compile and prepare a combined glacier inventory dataset for High Mountain Asia using Randolph Glacier Inventory(RGI) Regions 13,14 and15.
- To analyze the spatial distribution of glaciers across High Mountain Asia in terms of glacier count, glacier area, elevation and size-class characteristics.
- To compare glacier characteristics across Central Asia, South Asia West, and South Asia East.
- To inspect glacier mass-balance trends between 1990 and 2025 using time-series analysis and regional comparison.
- To create visualizations and dashboards that clearly communicate glacier patterns regional differences and glacier mass-loss trends.
- To evaluate forecasting models for predicting glacier mass balance using machine learning methods
- To establish a systematic analytical approach for glacier forecasting studies in the future under the influence of climate change.

1.4 Research Questions

- How have glacier area and elevation changed in the Himalayas over the past three decades (1990–2025)?
- Do glacier melting rates vary significantly across different sub-regions of the Himalayas (Western, Central, and Eastern)?
- Is the rate of glacier mass loss in the Himalayas accelerating over time?
- Which regions are most at risk of accelerated glacier loss?

2. LITERATURE REVIEW

2.1 Introduction

Glaciers across the Himalayas and wider Mountain Asia are experiencing significant retreat in recent years, reflecting the rising influence of climate change in the high altitude region. These glaciers are particularly important because Himalayan glaciers regulate the water supply for many people in that region. They function as fresh water-reservoirs and also influence glacier-related hazards. Research across High Mountain Asia demonstrates that the glacier mass balance varies over space and time, indicating that glacier response is not identical across the region. The variation shows the combined influence of climatic forcing, height, topography, debris cover and glacier shapes, all of which complicate the interpretation of long-term glacier change.

High Mountain Asia includes one of the most significant glacier systems outside the polar regions. A recent study demonstrates that Himalayan glaciers are mostly losing mass with some stable or even positive mass-balance behavior in some parts of Karakoram. So, the regional variability makes the glacier analysis complicated and the sub-regional investigation is required instead of considering the Himalayas as a single uniform glacier system. Thus, this review explores spatial variability, debris-covered glaciers, glacial lakes and GLOFs long term glacier changes, climate drivers, data scarcity and prediction methods.

2.2 Spatial Variability of Glacier Mass Loss

Spatial variability has emerged as one of the major themes in studies on Himalayan or High Mountain Asia (HMA) glaciers. Glacier mass does not follow a uniform pattern across the region because glacier response is influenced by climate, altitude, glacier size, glacier slope, aspect and regional topography. Glacier mass balance was estimated by Brun et al. (2017) from 2000 to 2016 for ~92% of the glacierized area in High Mountain Asia and was found to be -16.3 ± 3.5 Gt yr^{-1}. Their study also found that there were significant sub-regional differences, with negative mass balance areas including Nyainqentanglha and positive mass balance areas including Kunlun. This illustrates the fact that there is no single regional average for glacier change in HMA.

Within the Himalayan scale, the pattern of glacier change also differs from the Karakoram to western Himalaya, central Himalaya and eastern Himalaya. This difference is attributed to climatic systems differences, particularly the influence of the South Asian monsoon and the westerly winds, by Bolch et al. (2012). These differences affect snowfall, snow accumulation and snow melt throughout the region. In the same way, Patel et al. (2021) found a difference in glacier mass change between the different sub-regions of the Himalayas showing that single regional trend cannot used to explain the behaviour of glaciers. In addition, Robson et al. (2018) illustrate that glacier change differs according to the glacier mapping, altitude range and the distribution of supraglacial debris, with lower elevation glaciers resulting in greater mass loss especially in the Manaslu region.

Shean et al. (2020) created a systematic glacier mass balance regional assessment for High Mountain Asia with high-resolution WorldView and ASTER digital elevation models. The glacier mass loss estimated by their results is -19.0 ± 2.5 Gt yr^{-1} for the period 2000–2018, indicating a general mass loss in the HMA region, but also a high spatial variability between sub-regions and hydrological basins. This highlights that the importance of sub-regional analysis in the study of Himalayan glaciers

New studies based on altimetry also indicated that glacier are losing mass significantly. To compare glacier mass balance between 2003–2009 and 2018–2023, Hassan et al. (2026)

analyzed ICESat and ICESat-2 observations. They found glacier loss in the region has risen from 27.48 ± 7.96 Gt yr-1 to 36.58 ± 8.08 Gt yr-1, suggesting a greater glacier loss in High Mountain Asia during the recent warming and precipitation decrease. Moreover, Fan et al. (2023) estimated glacier mass balance over High Mountain Asia during 2000-2021, and found that the overall area had lost a lot of glacier mass, however, there were large differences among sub-regions, such as positive or stable glacier mass balance in certain areas.

2.3 Debris-Covered Glaciers

Debris cover plays a complex role in the glacier ablation and remains as the most concerned issue in Himalayan glacier studies. Racoviteanu et al. (2022) describe debris-covered glacier systems as particularly importance, as they affect the production of meltwater, glacier surface lowering, the formation of glacial lakes, and geomorphological hazards. While thick debris can decrease local surface melt by insulating the ice, satellite observations indicate that many debris-covered glacier tongues have very large surface lowering rates. This is partly due to the fact that ice cliffs, supraglacial ponds and complex glacier surface features may generate local melt “hot spots”.

In the past, large debris was considered to slow the melting of glaciers because of their insulating capacity at the surface of the glacier. However, recent studies show that the relationship between debris cover and glacier melt is more complicated. Chen et al. (2023) shows that spatial variability in melting on Himalayan debris-covered glaciers is controlled by thickness of debris and also by the presence of the ice cliffs and supraglacial ponds. In some areas, thick debris reduces melt at lower elevations, whereas the concentration of ice cliffs and ponds enhances localized ablation.

Robson et al. (2018) noted that debris- covered glaciers can undergo significant loss of ice even when frontal retreat is limited. It means that the frontal position of glacier is not reliable to indicate the health of the glacier. A glacier may look stable in length but still experiencing thinning and mass loss. Bolch et al. (2012) explained that debris cover and seasonal snow can make glacier mapping more difficult and change detection.

Overall, the literature explains that glacier cover does not prevent glaciers from melting. Rather, it alters the ablation process in various ways depending on debris thickness, surface morphology, slope, ice cliffs and supraglacial ponds. This is significant because glacier mass loss cannot be considered only through area or frontal retreat, elevation, surface conditions but regional setting must be considered.

2.4 Glacial Lakes and GLOFs

Glacial lakes are highly known as an important asset of a glacier change in the Himalayas. As glaciers melt and thinning can lead to the formation of the supraglacial, proglacial or moraine-dammed lakes. These lakes can increase ice loss by promoting thermal erosion, and ice-cliff retreat. Chen et al. (2023) identify supraglacial ponds and ice cliffs have key control on ablation over debris-covered ones, indicating that lake and pond expansion can increase localized melt.

Glacier lake expansion is related with glacial lake outburst flood(GLOF) hazards. Shrestha et al.(2007) show that glacier recession is often linked with the glacial lake expansion and increased glacial lake outburst flood(GLOF) hazard potential. GLOFs can create severe risk for the downstream communities, infrastructure and hydropower systems. Bajracharya and Mool(2009) inspected glaciers, glacial lake, and glacial lake outburst floods in the Mount Everest region of Nepal. They show that glacier retreat has promote the growth of glacial lakes, increasing the risk of glacial lake outburst floods and creating danger for the downstream communities and infrastructure .Quincey et al. (2007) also showed the use of remote sensing could be useful for early warning of glacial lake hazards.

Racoviteanu et al. (2022) demonstrate that the debris-covered glacier mass loss is often connected with the development of ice-contact and moraine-dammed lakes. Lakes are significant in these because they can modify the glacial melt processes, and also can enhance hazard potential downstream. The findings also indicate that moraine-dammed lakes are a future end-member scenario for debris-covered glaciers when warming continues.

The relationship between the glacial retreat and glacial lake hazard in Everest region is also explained by Shea et al. (2015). Their research shows that this quick retreat of terminuses can result in formation of proglacial lakes backed up by lateral and terminal moraines. Such, lakes can give rise to glacial lake outburst floods, posing a significant danger to downstream communities, infrastructure and hydropower systems. This indicate that glacier mass loss should not be viewed as an environmental issue but also as a hazard and water-resource management problem.

Therefore, understanding glacier mass loss needs more than measuring glacier area or retreat. High-risk glaciers need to measure mass balance more often with the proper understanding of lake development, glacier surface characteristics and downstream exposure. This is especially significant in debris covered glacier systems where low surface gradients and glacier velocities may increase the potential for moraine damming and the associated hazards.

2.5 Long-Term Trends

Long-term studies and observation confirm the increasing trend of glacier shrinking in the Himalayas region in recent decades. Kulkarni et al. (2007) examined an overall reduction in the glacier mass from 2077 km to 1628 km square by 2001 km square across selected Himalayan regions equivalent to 21 percent deglaciation. Also, recent regional assessments also report negative mass balance across the Himalayas, although the magnitude varies across sub-region (Patel et al., 2021).

Maurer et al. (2019) demonstrate the strong evidence that Himalayan ice loss has accelerated. Their study showed that the average rate of ice loss in 2000-2016 was almost double the rate during 1975-2000. It indicates that glacier mass loss is not only continuing but strengthening over time. In the Manaslu region, Robson et al. (2018) found average glacier lowering of 0.25 ± 0.08 m a^{-1} between 2000 and 2013, together with declining velocity and area. Therefore, it concludes that long term glacier shrinkage is widespread across the Himalayas, even though local rates of change substantially.

At a global scale, Hugonnet et al. (2021) found that the global glacier mass loss was found to be boosted during 2000-2019 and global glacier mass loss increased in the early twenty first century. It provides an important wider context for Himalayan glacier loss. Fan et al. (2023) also confirmed that mass loss in High Mountain Asia was substantial in 2000-21, but with regional differences. There studies confirm the necessity of the time-series mass balance analysis and forecasting of glaciers.

2.6 Climate Drivers of Glacier Mass Loss

Climate change is recognized as a main cause of glacier retreat, with increasing air temperatures, and changing precipitation patterns playing key roles. Wiltshire (2014) explains that the Hindu Kush-Karakoram-Himalaya region is affected by the major circulation systems that create strong east-west gradients in precipitation and snowfall seasonality.

Bolch et al. (2012) also noted that western part of the region is more affected by the westerlies, while the central and eastern Himalayas are more influenced by the summer monsoon. It is important because in monsoon dominated areas, mainly glaciers receive their more snowfall during summer along with melting is also high. Therefore, climate change can degrade the proportion of precipitation falling as snow and increase ablation.

Patel et al. (2021) report that regional mass change is strongly related with the snowfall, while this relationship weakens where runoff, evaporation or radiation exert stronger influence. Sherpa and Werth (2025) also describes that the precipitation seasonality is a key factor to control glacier mass across High Mountain Asia, especially in monsoon dominated glacier regions. Overall, the glacier loss is determined not only by warming but by precipitation timing, snowfall patterns and broader regional climate variability.

2.7 Methods and Data Limitations

Despite the major changes in glacier research, it is more difficult to accurately quantify glacier change in the Himalayan region. The main limitation is the scarcity of the field-based observation due to the remote place and high-altitude terrains. Robson et al. (2018) report that the in-situ glacier mass records are limited and biased toward accessible, small, and debris-free glaciers, making regional generalization difficult.

Therefore, due to such limitations, remote sensing has become the main device to monitor the glacier change. Satellite-based methods allow researchers to study glacial area, elevation change, velocity, and mass balance over the years. Local glacier mass-balance observations are rare in space and time particularly in the western part of High Mountain Asia, as Brun et al. (2017) explain, which makes the remote sensing methods such as DEM differencing, satellite altimetry, and gravimetry essential for regional-scale glacier assessment.

However, remote sensing based analysis also includes some uncertainties. These includes cloud cover, snow-shadow confusion, DEM errors, image quality limitations, and the difficulty of delineating debris covered glaciers. Fan et al used ICESat-2 and NASADEM to estimate glacier mass balance across High Mountain Asia indicating the value of satellite-based geodetic methods but radar penetration, spatial coverage and data uncertainty can affect the mass balance overall .

Shean et al. (2020) also stress that the various remote sensing approaches can yield varying mass-balance estimates owing to their sampling strategy, spatial resolution and sensitivity. This is why some of the earlier estimates of glacier mass loss in the HMA are not completely comparable. Hence it is important to merge several data sets and use well-defined regional boundaries when assessing glacier mass-balance data with greater reliability.

Remote sensing is also useful for glacier hazard assessment. Quincey et al. (2007) showed that satellite-based methods are useful to find potential glacial lake hazards in the remote mountain regions. However, it has its drawbacks such as glacier mapping and estimation of glacier mass balance can be impacted by cloud cover, snow-shadow confusion, DEM errors, radar penetration, debris covered ice and inconsistent temporal coverage. In addition, Cogley (2011) suggests the need for more advanced glacier inventories and improved analytical approaches to improve future predictions.

Overall, the study shows that both field-based and remote sensing methods have some difficulties. Field observations are good for making direct measurements but are limited to small area, whereas remote sensing can cover a large area but contains some uncertainties. Therefore, it is crucial to choose data carefully, use consistent regional boundaries, uncertainty awareness, and use combination of multiple data sources in possible.

2.8 Glacier Dynamics and Complexity

The literature presents Himalayan glacier as a complex and non-uniform as it depends upon the interactions between topography, elevation, accumulation regime, debris thickness ,ice cliffs , supraglacial lakes and local climate conditions. Robson et al. (2018) shows that glacier losses change according to the hypsometry and debris distribution, while Chen et al. (2023) demonstrate that various melt controls operate in different parts of debris-covered glaciers.

Kargel et al. (2011) demonstrate the Himalayan glacier system as a montage of diverse responses rather than a single coherent regional pattern. These studies have a common warming background, where individual's glacier and subs-regions can respond differently. Therefore, regional scale studies consider its complexity rather than assuming uniform glacier retreat across the Himalayas.

Bolch et al. (2012) also pointed out the uncertainty is due to the influence of extreme topographic relief, different climate and limited field observations in the Himalayan glaciers. So, glacier forecasting should be interpreted carefully. All physical glacier processes cannot be completely represented by a data-driven model unless climate, topographic and surface variables are included.

2.9 Predictive Modelling and Machine Learning Approaches

Many studies have been carried out about the glacier shrinking but fewer have focused on the predictive modelling of glacier mass balance in the Himalayas and High Mountain Asia. Recent research has begun to address this gap through machine learning and deep learning approaches.

Baraka et al. (2020) presents the value of machine learning for glacier monitoring and semi-automated glacier mapping in the Hindu Kush Himalaya region as old manually approaches are slow and require more manpower. Ren et al. (2024) compares the several machine learning models for simulating glacier mass balance in High Mountain Asia. Similarly, Gurung et al. (2025) also use machine learning models to predict the mass balance in the central Himalaya and report that extreme gradient boosting produces better results than other models.

Machine-learning models are useful because glacier mass balance is affected by complex and non-linear relationships between climate, elevation, glacier area, accumulation and ablation variables, implementing machine learning models can be fruitful. Models such as Random Forest and XGBoost are more useful to capture non-linear patterns than other linear methods. So, Van der Meer et al. (2025) introduced a miniML-MB, a machine learning mass-balance model based on a XGBoost, which is used to evaluate annual point surface mass balance using small datasets. They show that mean air temperature May to August and total precipitation October to February most strongly influences point mass balance. The model gave a better mean absolute error of 0.417 m w.e., which performed better than a simple positive degree-day model.

However, machine learning models also have drawbacks. Their performance depends upon on the quality, completeness and representatives of the input data. The Model cannot capture the physical causes of glacier change, if important climate variables such as temperature, precipitation and snowfall are missing. Also, Van der Meer et al. (2025) warned us that machine-learning models might not be accurate if the future or extreme conditions are outside of the range of the training set and their performance depends on the quality, completeness, and representativeness of the input data.

Therefore, machine learning is excellent for forecasting-oriented glacier analysis, especially for identifying patterns and predict future mass-balance trends but the results should be regarded as data-driven rather than complete physical simulations. So, machine learning is used to support mass-balance prediction, although there are some limitations on the available datasets.

2.10 Summary and Research Gap

Overall, the literature illustrates that Himalayan and High Mountain Asia glaciers are experiencing substantial mass retreat, primarily driven by rising temperature and changing precipitation patterns. However regional differences are influenced by elevation, topography, debris cover, and local climatic conditions. Research on long-term trends indicate that glacier mass loss has increased in recent decades and global and regional assessments show that glacial loss is an environmental and hydrological problem of significance.

The monitoring of glacier elevation changes and mass balance has been improved through remote sensing, but several uncertainties remain. These includes limited field observations, methodological differences, DEM errors, radar penetration, cloud cover, snow-shadow confusion and difficulties in mapping debris-covered glaciers. The literature also demonstrated that debris covered glaciers, supraglacial ponds and glacial lakes make glacier response more complex. Glacial lakes are very particularly significant as a connection between glacier retreat, GLOF hazards and downstream.

Existing studies have generated valuable information on glacier retreat, mass balance trends and the role of remote sensing but, yet important limitations still remain. In particular, across the Himalayan region, long term observations are uneven, field validation is sparse, and differences in method which make regional comparison difficult. Moreover, machine learning methods have become popular in glacier research, but few studies combine glacier inventory analysis, mass balance trends, regional comparison and modelling for glacier forecasting for sub-regions of the Himalayas.

Therefore, this study addresses an important gap by combining glacier-inventory analysis, mass balance trend analysis, regional comparison and machine learning-based forecasting. The method supports a data-driven modelling to examine historical mass loss across the

Himalayas and other parts of High Mountain Asia and provides an initiation for identifying regions that might be at higher risk of glacier loss.

3. METHODOLOGY

3.1 Research design and CRISP-DM Framework

This research was done to examine and predict glacier mass loss in the Himalayas and broader region of the High Mountain Asia region using a data-driven approach. The methodology is a combination of glacier inventory data, climate-related variables, historical glacial mass-balance, regional comparison and time-series forecasting. This methodology is appropriate because glacier change is affected by spatial factors such as glacier area, elevation, slope, region, as well as other factors such as annual mass balance and long-term climate-related change.

CRISP-DM stands for Cross-Industry Standard Process for Data Mining and this study follows the CRISP-DM framework because it offers a sequential and structural process for data analytics and machine learning projects. The stages of business understanding, data understanding, data preparation, modelling, evaluation, and deployment are reflected in this dissertation. In this study, business understanding is represented by environmental problem of glacier mass loss which impacts on water security and hazard risk. Data understanding and preparation are addressed through using RGI glacier inventory data and glacier mass-balance data. Time-series forecasting and regression-based prediction are used for modelling process, and MAE, RMSE and R square are used for evaluation stage. The final analysis stage assists in the identification of glacier trends, regional differences and high-risk glacier regions.

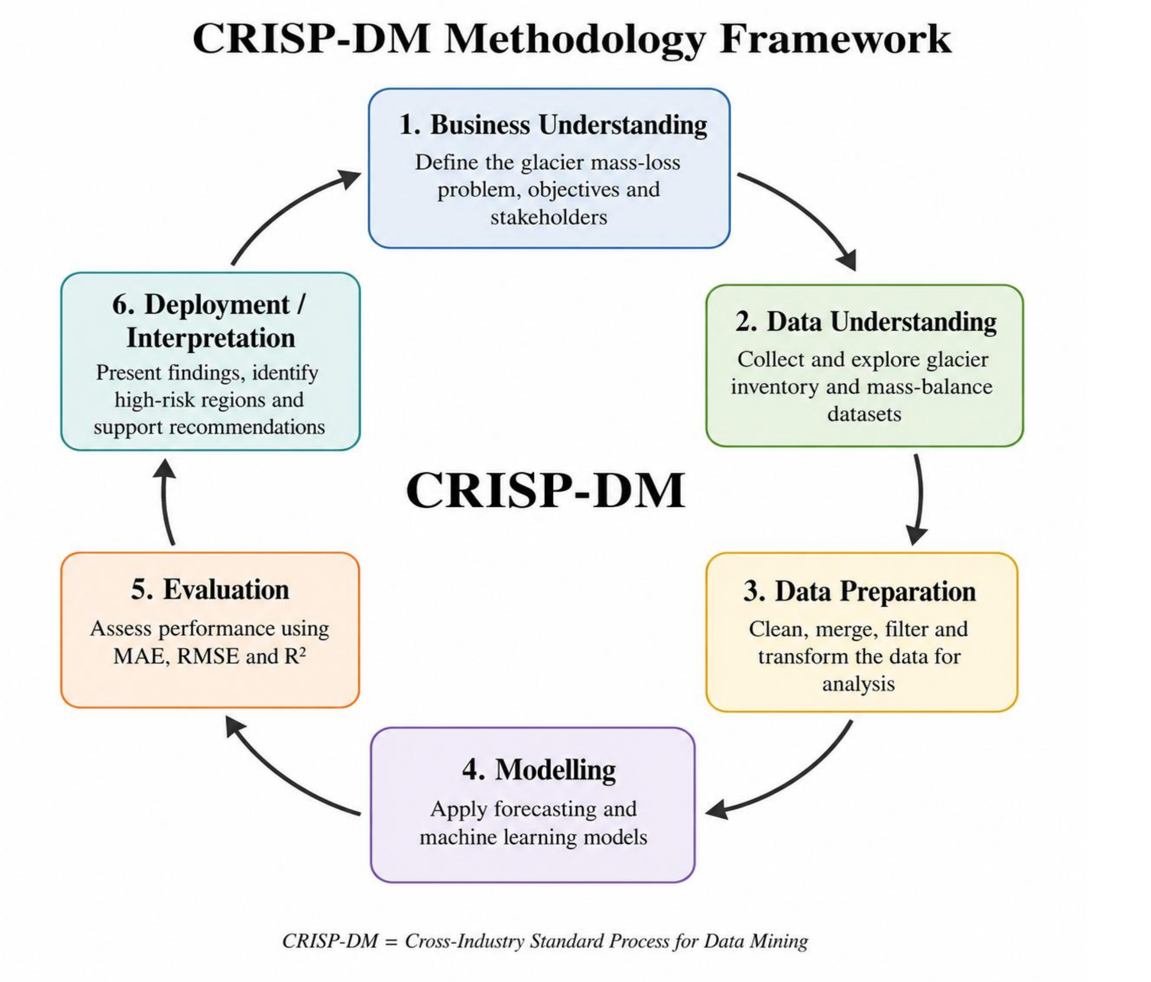

Figure 1: CRISP-DM methodology framework

The research follows a quantitative analytical design. First, to analyze the spatial features of glaciers in the High Mountain Asia, the data of glacial inventory was utilized. Then, mass-balance data of glaciers in the past were analyzed to determine the trends of mass-losses of the glaciers over time. At last, a regional comparison was used to test the hypothesis of whether there are differences in glacier loss across Himalayan sub-regions. Lastly, the forecasting methods were employed to determine whether the past glacier mass-balance trends could be used in the future to do so.

This method helps to address the main research questions of the dissertation, which are concerned with the glacier area and height change, the regional difference in the glacier melting, the acceleration of mass loss over time and the identification of high-risk glacier regions.

3.2 Study Area

This study covers both the Himalayas and wider High Mountains Asia, which is one of the most significant glacierised areas outside the polar region. High Mountain Asia has some of the largest mountain ranges that include the Himalayas, Hindu Kush, Karakoram, Pamir and Tibetan Plateau. These glacier systems are valuable in that they supply the downstream communities, agriculture, hydropower and ecosystems with freshwater resources.

To conduct the study on spatial glacier inventory analysis, Randolph Glacier inventory regions used in the study are:

- Region 13: Central Asia
- Region 14: South Asia
- Region 15: South East Asia

These regions are chosen as they include major glacier systems connected to the Himalayas and surrounding high-mountain environments. For the time-series mass-balance analysis, the study focused on Himalayan and High Mountain Asia-related countries available in the mass-balance dataset, including India, Nepal, Bhutan and China. The countries were used as local indicators to the Western Himalaya, Central Himalaya, Eastern Himalaya and the greater region of High Mountain Asia/Tibetan region.

3.3 Data Sources

Two major forms of secondary data were used in this study. First one is glacier inventory data and second one is glacier mass-balance data. The data of the glaciers inventory were analyzed spatially and descriptively, whereas the data of the mass-balance were analyzed by the time-series analysis and by the prediction.

Dataset	Source	Main Use
RGI Region 13	Randolph Glacier Inventory	Central Asia glacier inventory analysis
RGI Region 14	Randolph Glacier Inventory	South Asia West glacier inventory analysis
RGI Region 15	Randolph Glacier Inventory	South Asia East glacier inventory analysis
Mass-balance dataset	Secondary glacier mass-balance data	Time-series and forecasting analysis

Figure 2: Overview of datasets used in the study

3.3.1 Glacier Inventory Data

The primary source for the data is the Randolph Glacier Inventory (RGI) v7.0. The following RGI region was selected:

- Region 13: Central Asia
- Region 14 : South Asia West
- Region 15 : South Asia East

The datasets for the Central Asia, South Asia West and South Asia West are as given below:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	rgi_id	o1region	o2region	glims_id	anlys_id	subm_id	src_date	cenlon	cenlat	utm_zone	area_km2	primeclas	conn_lvl	surge_typ
2	RGI2000-v	13	13-01	G067426E	804440	752	2002-07-1	67.42588	38.74331	42	0.011076	0	0	0
3	RGI2000-v	13	13-01	G067480E	804446	752	2002-07-1	67.47962	38.71458	42	0.014615	0	0	0
4	RGI2000-v	13	13-01	G067485E	804448	752	2002-07-1	67.48497	38.71343	42	0.013399	0	0	0
5	RGI2000-v	13	13-01	G067489E	804451	752	2002-07-1	67.48941	38.71449	42	0.014232	0	0	0
6	RGI2000-v	13	13-01	G067492E	804453	752	2002-07-1	67.49194	38.71371	42	0.048107	0	0	0
7	RGI2000-v	13	13-01	G067499E	804457	752	2002-07-1	67.49871	38.70956	42	0.011037	0	0	0
8	RGI2000-v	13	13-01	G067501E	804458	752	2002-07-1	67.50067	38.71415	42	0.015431	0	0	0
9	RGI2000-v	13	13-01	G067507E	804463	752	2002-07-1	67.50654	38.71119	42	0.1126	0	0	0
10	RGI2000-v	13	13-01	G067503E	804460	752	2002-07-1	67.5033	38.68639	42	0.021668	0	0	0
11	RGI2000-v	13	13-01	G067509E	804465	752	2002-07-1	67.5084	38.68103	42	0.025638	0	0	0
12	RGI2000-v	13	13-01	G067514E	804471	752	2002-07-1	67.51449	38.68233	42	0.02473	0	0	0
13	RGI2000-v	13	13-01	G067507E	804462	752	2002-07-1	67.50673	38.66166	42	0.022496	0	0	0
14	RGI2000-v	13	13-01	G067504E	804461	752	2002-07-1	67.50465	38.65673	42	0.032482	0	0	0
15	RGI2000-v	13	13-01	G067508E	804464	752	2002-07-1	67.50754	38.65182	42	0.033806	0	0	0
16	RGI2000-v	13	13-01	G067509E	804466	752	2002-07-1	67.50871	38.64523	42	0.033863	0	0	0
17	RGI2000-v	13	13-01	G067516E	804472	752	2002-07-1	67.51585	38.6466	42	0.323903	0	0	0
18	RGI2000-v	13	13-01	G067530E	804480	752	2002-07-1	67.52989	38.63125	42	0.013461	0	0	0
19	RGI2000-v	13	13-01	G067545E	804488	752	2002-07-1	67.54491	38.69206	42	0.018618	0	0	0
20	RGI2000-v	13	13-01	G067555E	804495	752	2002-07-1	67.55502	38.69002	42	0.04015	0	0	0
21	RGI2000-v	13	13-01	G067534E	804482	752	2002-07-1	67.53408	38.73666	42	0.033734	0	0	0
22	RGI2000-v	13	13-01	G067524E	804476	752	2002-07-1	67.52397	38.73858	42	0.084744	0	0	0
23	RGI2000-v	13	13-01	G067544E	804487	752	2002-07-1	67.54441	38.75905	42	0.051744	0	0	0
24	RGI2000-v	13	13-01	G067548E	804489	752	2002-07-1	67.54768	38.76056	42	0.015485	0	0	0
25	RGI2000-v	13	13-01	G067552E	804492	752	2002-07-1	67.55211	38.76034	42	0.045405	0	0	0
26	RGI2000-v	13	13-01	G067556E	804496	752	2002-07-1	67.55575	38.7591	42	0.077115	0	0	0

	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB
1	term_typ	glac_nam	is_rgi6	termlon	termlat	zmin_m	zmax_m	zmed_m	zmean_m	slope_de	aspect_de	aspect_se	dem_sour	lmax_m
2	9		0	67.42658	38.74383	3693.856	3783.966	3727.242	3728.608	34.00393	26.57254	2	COPDEM3	151
3	9		0	67.47868	38.71486	3839.787	3899.867	3873.05	3870.65	29.50686	344.1754	1	COPDEM3	181
4	9		0	67.48534	38.71382	3844.892	3897.4	3862.385	3864.616	24.49718	357.8179	1	COPDEM3	107
5	9		0	67.49037	38.71517	3707.877	3772.552	3734.059	3735.828	22.8099	18.29015	1	COPDEM3	181
6	9		0	67.49068	38.71398	3750.389	3957.731	3809.629	3825.5	44.84194	333.275	8	COPDEM3	236
7	9		0	67.498	38.71003	3947.011	3990.163	3976.599	3973.924	17.21633	330.595	8	COPDEM3	198
8	9		0	67.50136	38.71481	3796.266	3899.976	3836.74	3838.795	32.81758	24.1156	2	COPDEM3	178
9	9		0	67.50601	38.71287	3730.04	3865.484	3785.632	3787.255	23.12626	11.97782	1	COPDEM3	471
10	9		0	67.50439	38.68671	3982.295	4030.842	4015.036	4012.624	17.60729	40.83388	2	COPDEM3	201
11	9		0	67.51011	38.68213	3814.317	3972.813	3866.669	3872.837	28.39384	357.7934	1	COPDEM3	313
12	9		0	67.51419	38.68299	3735.481	3836.981	3780.6	3782.802	36.10575	349.8361	1	COPDEM3	253
13	9		0	67.50726	38.66237	3800.558	3903.985	3848.411	3851.19	35.26147	2.710188	1	COPDEM3	186
14	9		0	67.50552	38.65711	3875.418	3996.709	3900.339	3908.293	23.77809	38.93653	2	COPDEM3	247
15	9		0	67.50917	38.65224	3882.981	4040.675	3921.914	3930.327	25.21828	44.92584	2	COPDEM3	303
16	9		0	67.50891	38.64626	3905.474	3993.085	3929.726	3934.623	21.65358	30.36611	2	COPDEM3	259
17	9		0	67.51584	38.64996	3710.671	3958.055	3810.938	3813.219	19.74318	351.733	1	COPDEM3	898
18	9		0	67.53031	38.63207	3595.599	3669.383	3632.537	3631.729	26.92553	52.46849	2	COPDEM3	192
19	9		0	67.54522	38.69246	3789.598	3865.644	3818.368	3820.34	32.07084	8.582625	1	COPDEM3	114
20	9		0	67.55508	38.69113	3795.746	3905.808	3836.775	3839.723	26.78159	340.2742	1	COPDEM3	203
21	9		0	67.53482	38.73777	3710.406	3799.475	3753.2	3753.445	27.79412	341.5119	1	COPDEM3	351
22	9		0	67.52485	38.74019	3832.131	3982.931	3883.878	3893.26	23.78192	354.484	1	COPDEM3	457
23	9		0	67.54581	38.76063	3534.091	3773.526	3616.188	3626.262	32.63568	341.139	1	COPDEM3	359
24	9		0	67.54787	38.76114	3491.942	3670.141	3555.561	3565.295	53.58486	341.9257	1	COPDEM3	123
25	9		0	67.55318	38.76178	3425.545	3674.275	3537.119	3535.572	37.76139	13.95287	1	COPDEM3	365
26	9		0	67.55739	38.76108	3401.795	3656.58	3491.257	3503.67	32.03883	16.38019	1	COPDEM3	470

Figure 3: Dataset for Central Asia

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	rgi_id	o1region	o2region	glims_id	anlys_id	subm_id	src_date	cenlon	cenlat	utm_zone	area_km2	primeclass	conn_lvl	surge_typ
2	RGI2000-v	14	14-01	G067620E3	804535	752	2002-07-1	67.6202	34.65491	42	0.054023	0	0	0
3	RGI2000-v	14	14-01	G067622E3	804537	752	2002-07-1	67.62191	34.65106	42	0.031114	0	0	0
4	RGI2000-v	14	14-01	G067628E3	804541	752	2002-07-1	67.62778	34.64906	42	0.167697	0	0	0
5	RGI2000-v	14	14-01	G067631E3	804550	752	2002-07-1	67.63272	34.64667	42	0.263019	0	0	0
6	RGI2000-v	14	14-01	G067639E3	804560	752	2002-07-1	67.63896	34.65432	42	0.035827	0	0	0
7	RGI2000-v	14	14-01	G067643E3	804562	752	2002-07-1	67.64345	34.64384	42	0.158453	0	0	0
8	RGI2000-v	14	14-01	G067650E3	804566	752	2002-07-1	67.65035	34.64584	42	0.113578	0	0	0
9	RGI2000-v	14	14-01	G067657E3	804570	752	2002-07-1	67.65636	34.64425	42	0.053281	0	0	0
10	RGI2000-v	14	14-01	G067663E3	804573	752	2002-07-1	67.66338	34.64243	42	0.124505	0	0	0
11	RGI2000-v	14	14-01	G067670E3	804576	752	2002-07-1	67.66953	34.63765	42	0.099842	0	0	0
12	RGI2000-v	14	14-01	G067678E3	804579	752	2002-07-1	67.67741	34.63848	42	0.053151	0	0	0
13	RGI2000-v	14	14-01	G067684E3	804588	752	2002-07-1	67.68447	34.64415	42	0.020362	0	0	0
14	RGI2000-v	14	14-01	G067689E3	804597	752	2002-07-1	67.68961	34.65001	42	0.282738	0	0	0
15	RGI2000-v	14	14-01	G067696E3	804600	752	2002-07-1	67.69589	34.65121	42	0.103261	0	0	0
16	RGI2000-v	14	14-01	G067704E3	804606	752	2002-07-1	67.70362	34.6502	42	0.020884	0	0	0
17	RGI2000-v	14	14-01	G067712E3	804610	752	2002-07-1	67.71199	34.65411	42	0.029208	0	0	0
18	RGI2000-v	14	14-01	G067727E3	804613	752	2002-07-1	67.72745	34.64947	42	0.058118	0	0	0
19	RGI2000-v	14	14-01	G067735E3	804618	752	2002-07-1	67.73451	34.64946	42	0.075841	0	0	0
20	RGI2000-v	14	14-01	G067747E3	804630	752	2002-07-1	67.74748	34.64726	42	0.100949	0	0	0
21	RGI2000-v	14	14-01	G067761E3	804639	752	2002-07-1	67.76078	34.6482	42	0.073733	0	0	0
22	RGI2000-v	14	14-01	G067767E3	804647	752	2002-07-1	67.76669	34.65013	42	0.035063	0	0	0
23	RGI2000-v	14	14-01	G067774E3	804653	752	2002-07-1	67.77522	34.6458	42	0.118163	0	0	0
24	RGI2000-v	14	14-01	G067842E3	804683	752	2002-07-1	67.84238	34.62636	42	0.046035	0	0	0
25	RGI2000-v	14	14-01	G067870E3	804707	752	2002-07-1	67.87043	34.61359	42	0.047966	0	0	0

	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB
1	term_type	glac_name	is_rgi6	termlon	termlat	zmin_m	zmax_m	zmed_m	zmean_m	slope_deg	aspect_de	aspect_se	dem_sour	lmax_m
2	9		0	67.62193	34.65582	4536.712	4676.8	4603.528	4603.859	26.94893	37.6888	2	COPDEM3	366
3	9		0	67.62336	34.65183	4638.783	4804.813	4717.56	4717.868	32.96445	52.13326	2	COPDEM3	265
4	9		0	67.63026	34.65127	4365.106	4679.718	4466.458	4474.748	28.8386	51.14768	2	COPDEM3	536
5	9		0	67.63194	34.65099	4355.62	4613.721	4466.956	4467.86	18.51209	356.5536	1	COPDEM3	1040
6	9		0	67.63942	34.65541	4330.189	4459.232	4376.481	4382.614	31.87757	6.774054	1	COPDEM3	258
7	9		0	67.6445	34.64662	4447.804	4612.23	4521.74	4521.17	18.5295	0.267812	1	COPDEM3	668
8	9		0	67.65121	34.64824	4535.062	4692.041	4599.747	4601.608	19.94552	334.0143	8	COPDEM3	541
9	9		0	67.65878	34.64505	4396.76	4583.117	4475.795	4476.6	28.98005	65.41449	2	COPDEM3	346
10	9		0	67.66524	34.64454	4225.439	4507.302	4311.571	4319.856	26.13405	1.384231	1	COPDEM3	530
11	9		0	67.66989	34.63924	4368.482	4501.345	4411.879	4414.609	22.34325	11.63736	1	COPDEM3	397
12	9		0	67.67681	34.63927	4472.042	4578.994	4495.887	4503.077	22.64942	345.3795	1	COPDEM3	252
13	9		0	67.68309	34.64472	4643.528	4789.776	4697.193	4705.029	29.78447	318.5782	8	COPDEM3	292
14	9		0	67.68479	34.65318	4215.793	4705.971	4386.956	4384.172	24.52928	321.2761	8	COPDEM3	1038
15	9		0	67.69464	34.65266	4453.619	4642.565	4524.277	4527.775	26.86188	322.1858	8	COPDEM3	450
16	9		0	67.703	34.65019	4547.612	4587.438	4564.161	4565.062	25.11071	321.2626	8	COPDEM3	104
17	9		0	67.71204	34.65472	4236.273	4321.006	4260.791	4265.345	22.90933	348.7962	1	COPDEM3	213
18	9		0	67.726	34.65059	4373.345	4477.703	4416.459	4418.093	23.78788	343.0819	1	COPDEM3	321
19	9		0	67.73494	34.65106	4358.256	4512.168	4397.254	4403.506	21.62946	341.4213	1	COPDEM3	384
20	9		0	67.74705	34.64847	4353.34	4487.025	4403.625	4404.018	23.05952	355.0887	1	COPDEM3	416
21	9		0	67.76061	34.64921	4300.211	4388.148	4331.765	4334.036	21.72467	350.6658	1	COPDEM3	288
22	9		0	67.76571	34.65109	4327.555	4511.534	4405.264	4410.575	41.6169	308.7873	8	COPDEM3	265
23	9		0	67.7768	34.64727	4342.684	4457.781	4393.824	4393.354	21.36949	358.176	1	COPDEM3	603
24	9		0	67.84325	34.62738	4201.351	4307.699	4245.665	4246.148	21.8228	350.7748	1	COPDEM3	251
25	9		0	67.87162	34.61385	4291.312	4442.82	4337.28	4342.222	33.63033	49.21184	2	COPDEM3	268

Figure 4: Dataset for South Asia West

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	rgi_id	o1region	o2region	glims_id	anlys_id	subm_id	src_date	cenlon	cenlat	utm_zone	area_km2	primeclas	conn_lvl	surge_typ
2	RGI2000-v	15	15-01	G078088E3	866850	752	2002-07-1(	78.08789	31.39805	44	0.032645	0	0	0
3	RGI2000-v	15	15-01	G078125E3	867227	752	2002-07-1(	78.1237	31.3978	44	0.26063	0	0	0
4	RGI2000-v	15	15-01	G078128E3	867273	752	2000-08-0!	78.12851	31.39029	44	0.040742	0	0	0
5	RGI2000-v	15	15-01	G078161E3	867575	752	2000-08-0!	78.16068	31.3759	44	0.152257	0	0	0
6	RGI2000-v	15	15-01	G078151E3	867482	752	2000-08-0!	78.14975	31.36399	44	0.278587	0	0	0
7	RGI2000-v	15	15-01	G078137E3	867352	752	2000-08-0!	78.1371	31.35499	44	0.050078	0	0	0
8	RGI2000-v	15	15-01	G078163E3	867595	752	2000-08-0!	78.16334	31.34661	44	0.08874	0	0	0
9	RGI2000-v	15	15-01	G078270E3	868597	752	2000-08-0!	78.27039	31.27346	44	0.03915	0	0	0
10	RGI2000-v	15	15-01	G078278E3	868682	752	2000-08-0!	78.27715	31.26477	44	0.326227	0	0	0
11	RGI2000-v	15	15-01	G078295E3	868854	752	2000-08-0!	78.2946	31.26128	44	0.053387	0	0	0
12	RGI2000-v	15	15-01	G078301E3	868899	752	2000-08-0!	78.30023	31.26015	44	0.038328	0	0	0
13	RGI2000-v	15	15-01	G078305E3	868962	752	2000-08-0!	78.30596	31.26244	44	0.234542	0	0	0
14	RGI2000-v	15	15-01	G078313E3	869058	752	2000-08-0!	78.31409	31.27798	44	0.25082	0	0	0
15	RGI2000-v	15	15-01	G078306E3	869007	752	2000-08-0!	78.30985	31.28077	44	0.1986	0	0	0
16	RGI2000-v	15	15-01	G078303E3	868926	752	2000-08-0!	78.30267	31.28385	44	0.109161	0	0	0
17	RGI2000-v	15	15-01	G078307E3	868973	752	2000-08-0!	78.30705	31.29112	44	0.640818	0	0	0
18	RGI2000-v	15	15-01	G078290E3	868816	752	2000-08-0!	78.2916	31.29826	44	0.360023	0	0	0
19	RGI2000-v	15	15-01	G078292E3	868828	752	2000-08-0!	78.29327	31.30536	44	0.611464	0	0	0
20	RGI2000-v	15	15-01	G078299E3	868889	752	2002-07-1(	78.29913	31.3061	44	0.048894	0	0	0
21	RGI2000-v	15	15-01	G078294E3	868855	752	2000-08-0!	78.29524	31.31335	44	0.184202	0	0	0
22	RGI2000-v	15	15-01	G078322E3	869144	752	2000-08-0!	78.32258	31.27753	44	0.074487	0	0	0
23	RGI2000-v	15	15-01	G078327E3	869185	752	2000-08-0!	78.32661	31.26987	44	0.844862	0	0	0
24	RGI2000-v	15	15-01	G078337E3	869302	752	2000-08-0!	78.33697	31.27383	44	0.660524	0	0	0
25	RGI2000-v	15	15-01	G078350E3	869429	752	2002-07-1(	78.34897	31.27311	44	0.222631	0	0	0

	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB
1	term_type	glac_name	is_rgi6	termlon	termlat	zmin_m	zmax_m	zmed_m	zmean_m	slope_deg	aspect_de	aspect_se	dem_sour	lmax_m
2	9		0	78.08796	31.39741	4662.295	4699.21	4669.472	4671.425	13.42707	122.2673	4	COPDEM3	173
3	9		0	78.12104	31.39314	4453.358	4705.992	4570.947	4571.277	22.82298	269.6691	7	COPDEM3	1113
4	9		0	78.12852	31.38882	4791.759	4858.681	4832.184	4827.67	15.62626	212.7197	6	COPDEM3	327
5	9		0	78.15731	31.3754	4495.664	4760.434	4618.216	4623.046	25.98365	291.6897	7	COPDEM3	573
6	9		0	78.14962	31.3688	4249.437	4717.812	4477.985	4478.402	27.61453	338.1906	1	COPDEM3	1209
7	9		0	78.13556	31.35498	4741.931	4885.46	4841.555	4837.064	28.96891	308.3773	8	COPDEM3	253
8	9		0	78.163	31.34847	4618.841	4808.789	4698.913	4700.624	28.2299	342.7354	1	COPDEM3	461
9	9		0	78.26921	31.27398	4601.146	4722.705	4648.865	4651.769	31.08743	322.3659	8	COPDEM3	260
10	9		0	78.27123	31.26619	4605.71	4938.588	4791.702	4785.682	22.74241	297.6895	8	COPDEM3	1006
11	9		0	78.29431	31.26215	4671.062	4795.16	4728.388	4729.499	33.46623	355.061	1	COPDEM3	293
12	9		0	78.30009	31.26182	4659.037	4782.83	4707.585	4709.942	24.1014	345.9423	1	COPDEM3	322
13	9		0	78.30062	31.26404	4569.646	4865.677	4734.929	4726.891	22.01337	333.3874	8	COPDEM3	605
14	9		0	78.31146	31.27533	4864.22	5118.635	4964.848	4966.729	20.4793	222.7643	6	COPDEM3	825
15	9		0	78.30516	31.27958	4825.837	5214.065	5005.836	5007.835	26.88345	242.3008	6	COPDEM3	795
16	9		0	78.30206	31.28211	4961.343	5130.484	5011.247	5019.753	25.69033	257.5909	7	COPDEM3	506
17	9		0	78.30788	31.29772	4653.11	5269.536	4773.162	4837.417	22.73237	334.9095	8	COPDEM3	1687
18	9		0	78.2855	31.298	4643.739	5049.495	4928.024	4915.294	23.60823	290.9752	7	COPDEM3	803
19	9		0	78.28691	31.30884	4698.219	5167.936	4891.366	4903.776	20.78898	316.4093	8	COPDEM3	1299
20	9		0	78.30066	31.30616	4925.753	5021.679	4964.963	4967.084	26.67659	45.39683	2	COPDEM3	282
21	9		0	78.29114	31.31424	4849.234	5115.286	5001.107	4994.872	25.30209	259.4773	7	COPDEM3	633
22	9		0	78.32046	31.27549	4946.506	5089.223	5015.81	5013.09	21.92385	264.6579	7	COPDEM3	479
23	9		0	78.32131	31.2695	4744.619	5136.857	4844.955	4861.127	16.92068	281.1879	7	COPDEM3	1433
24	9		0	78.34257	31.2748	4844.879	5076.95	4942.747	4940.085	16.15081	121.089	4	COPDEM3	1133
25	9		0	78.34697	31.26981	4804.395	5133.546	4906.68	4919.739	20.58584	244.6507	6	COPDEM3	909

Figure 5: Dataset for South Asia East

These datasets consist of various attributes such as glacier identifier, geographic coordinates, glacier area, elevation range, median elevation, slope, aspect, maximum glacier length, and terminus type. The RGI datasets are selected because they provide a standardized and widely used glacier inventory framework for regional glacier analysis.

The RGI data was mainly used to analyze glacier count, total glacier area elevation distribution, size-class structure and regional differences across High Mountain Asia.

3.3.2 Glacier Mass-Balance Dataset

The second significant dataset that was used in this study was the glacier mass-balance dataset. This dataset has historical observations of the mass-balance of the glaciers in the form of annual values of the mass-balance, area of the glaciers, equilibrium line altitude, accumulation-area ratio and seasonal values of the mass-balance**.**

The main variable used for time-series analysis and forecasting was:

- balance

This variable represents the annual glacier mass-balance value and it was used as the key indicator of the mass loss of the glaciers. The negative balance values show that the glaciers are losing mass whereas positive values show that the glaciers are gaining mass. The dataset for the glacier mass-balance dataset are as given below:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	country	glacier_name	glacier_id	year	id	original_i	time_syst	begin_dat	begin_dat	end_date	end_date	latitude	longitude	elevation	balance	balance_u	density	density_u	method	balance_c	remarks
2	AF	PIR YAKH	10452	2018	40809	PY10		7/7/2017	0	#######	0	35.5987	70.18	4541	-3.413		910			index	
3	AF	PIR YAKH	10452	2018	40810	PY20		7/7/2017	0	#######	0	35.6008	70.1742	4737	-1.684		910			index	
4	AQ	BAHIA DEL DIA	2665	2010	36568			3/1/2009	0	#######	0			637	0.3					annual	
5	AQ	BAHIA DEL DIA	2665	2010	36564			3/1/2009	0	#######	0			551	0.6					annual	
6	AQ	BAHIA DEL DIA	2665	2010	36563			3/1/2009	0	#######	0			515	0.6					annual	
7	AQ	BAHIA DEL DIA	2665	2010	36562			3/1/2009	0	#######	0			505	0.6					annual	
8	AQ	BAHIA DEL DIA	2665	2010	36576			3/1/2009	0	#######	0			288	0.4					annual	
9	AQ	BAHIA DEL DIA	2665	2010	36577			3/1/2009	0	#######	0			375	0.2					annual	
10	AQ	BAHIA DEL DIA	2665	2010	36561			3/1/2009	0	#######	0			445	1					annual	
11	AQ	BAHIA DEL DIA	2665	2010	36560			3/1/2009	0	#######	0			464	0.6					annual	
12	AQ	BAHIA DEL DIA	2665	2010	36559			3/1/2009	0	#######	0			458	0.4					annual	
13	AQ	BAHIA DEL DIA	2665	2010	36558			3/1/2009	0	#######	0			456	0.7					annual	
14	AQ	BAHIA DEL DIA	2665	2010	36557			3/1/2009	0	#######	0			442	0.6					annual	
15	AQ	BAHIA DEL DIA	2665	2010	36556			3/1/2009	0	#######	0			398	0.5					annual	
16	AQ	BAHIA DEL DIA	2665	2010	36555			3/1/2009	0	#######	0			100	0.1					annual	
17	AQ	BAHIA DEL DIA	2665	2010	36574			3/1/2009	0	#######	0			272	0.4					annual	
18	AQ	BAHIA DEL DIA	2665	2010	36575			3/1/2009	0	#######	0			273	0.2					annual	
19	AQ	BAHIA DEL DIA	2665	2010	36573			3/1/2009	0	#######	0			270	0.1					annual	
20	AQ	BAHIA DEL DIA	2665	2010	36572			3/1/2009	0	#######	0			205	0.1					annual	
21	AQ	BAHIA DEL DIA	2665	2010	36567			3/1/2009	0	#######	0			553	0.7					annual	
22	AQ	BAHIA DEL DIA	2665	2010	36566			3/1/2009	0	#######	0			115	0.2					annual	
23	AQ	BAHIA DEL DIA	2665	2010	36571			3/1/2009	0	#######	0			167	0.2					annual	
24	AQ	BAHIA DEL DIA	2665	2010	36565			3/1/2009	0	#######	0			588	0.3					annual	
25	AQ	BAHIA DEL DIA	2665	2010	36570			3/1/2009	0	#######	0			412	0.6					annual	

Figure 6: Dataset for Glacier Mass-Balance

The dataset also includes other useful variables such as:

- year
- glacier_name
- glacier_id
- country
- area
- ela
- aar
- winter_balance
- summer_balance

In this dissertation, the data was filtered down to the time frame between 1990 and 2025, which corresponds with the research proposal. The country variable was to assist in the regional comparison between the Himalayan and the High Mountain Asia related areas.

3.3.3 Supporting Literature and Secondary sources

In order to support the interpretation of glacier mass loss, regional glacier behavior, debris-covered glaciers, glacial lakes, climate drivers and data limitations, published research papers were used. Previous studies were carried out to explain why glacier mass loss varies across the Himalayas and why regional analysis is needed.

3.4 Data Preparation and Processing

Data preparation was performed using Power BI, Excel and Python. Power BI was mainly used for data cleaning, creating dashboard and visual analysis. Python was involved for time-series analysis, preparing forecasting and statistical modelling.

The data preparation consists of several steps.

Firstly, the three RGI regional attribute datasets were imported into Power Bi. A new variable name region was created for each dataset to identify the region as Central Asia, South Asia West and South Asia East. Then, the three tables were standardized by aligning column names and keeping only variables which are relevant to our analysis.

These cleaned RGI datasets were appended into one combined dataset named HMA_Glacier. This dataset was used for glacier inventory analysis and regional comparison.

For the mass-balance of the glaciers, dataset was imported and verified in terms of missing values, inconsistency in data types and duplicate records. The dataset was narrowed down to years 1990-2025 since the research questions were limited to the changes that occurred in glaciers over the past 30 years. Regional analysis was done by selecting countries that were relevant to the Himalayas and the High Mountain Asia.

Fourth, a new variable on a regional classification was created in a mass-balance dataset. The country codes were then clustered into larger sub-regions as follows:

- India = Western Himalaya
- Nepal = Central Himalaya
- Bhutan = Eastern Himalaya
- China = Tibetan HMA

The patterns of mass-balance of glaciers across various regions related to Himalayan were compared using this classification.

3.4.1 Data Cleaning and Standardization

To maintain the consistency and reliability in the both datasets, data cleaning was applied. Across the three regional RG1 datasets, data cleaning and standardization were applied which includes the inspecting for missing values, eliminating unnecessary columns, correcting inconsistent region labels, aligning variable names and checking data types.

Glacier names are missing in some parts of the RGI dataset. So, the glacier identifier rgi_id was used as the main reference field. Each glacier can be uniquely identified even where glacier names are unavailable.

For the mass-balance dataset, the year column was checked to ensure that it was in numerical format. The balance column was checked because it was the main target variable for analysis and forecasting. The values which are not available, were not included in the modelling dataset as they could be used for trend or prediction analysis.

3.4.2 Feature Engineering

Feature engineering was applied to improve the data more analytically.

For the RGI inventory dataset, the size_class variable was generated based on the glacier area to classify the glaciers by size which allowed for easier description and comparison between regions.

For the mass-balance dataset, a variable named sub_region was constructed from the country column. This allowed the study to compare the trends across the western Himalaya, central Himalaya, eastern Himalaya and the greater region of High Mountain Asia.

A decade variable was also created from the year column. This enables the comparison of the glacier mass balance across various time periods, including the 1990s, 2000s, 2010s and 2020, 2025. This was useful to determine whether the glacier mass loss has increased over time.

3.5 Variables Used in the Study

The main variables used from the RGI inventory dataset are as follows:

- glacier ID (rgi_id)
- central latitude and longitude
- glacier area (area_km2)
- minimum elevation (zmin_m)
- maximum elevation (zmax_m)
- median elevation (zmed_m)
- slope (slope_deg)
- aspect (aspect_deg)
- maximum glacier length(lmax_m)
- terminus type(term_type)
- region
- size class

The main variables from the mass-balance dataset were:

- year
- country
- sub-region
- glacier name
- glacier ID
- balance
- glacier area
- equilibrium line altitude
- accumulation-area ratio
- winter balance
- summer balance

The most important variable for forecasting was balance because it directly represents yearly glacier mass-balance change. Negative values were represented as glacier mass loss.

3.6 Data Analysis Methods

The analysis was conducted into three stages: descriptive, comparative regional analysis, and forecasting-oriented interpretation.

3.6.1 Descriptive Analysis

This analysis is used to summarize the main characteristics of glaciers across High Mountain Asia (HMA) which includes glacier counts, total glacier area, mean and median elevation, and the distribution of glacier sizes.

For the mass-balance dataset, the mean annual balance by year, country, sub-region, and decade was calculated by using descriptive analysis and also helps to identify the general patterns of glacier mass loss over time.

3.6.2 Comparative Regional Analysis

Comparison was conducted between Central Asia, South Asia West and South Asia East. It examined the number of glaciers, total glacier area, median elevation of glaciers and glacier size distribution

In the case of the mass-balance dataset, India, Bhutan, and China were divided into larger Himalayan-related sub-regions. Those sub-regions are compared on the annual values of balances. This helped to calculate the variation of glacier melting across the different regions.

This analysis was important because glacier mass loss is not spatially uniform. Past researchers show that glacier response varies according to the climate, elevation, topography, debris cover and regional precipitation patterns.

3.6.3 Time-Series Trend Analysis

This analysis was used to calculate the changed glacier mass balance during the 1990 and 2025. The main variable used in this analysis was balance.

To obtain the average annual mass balance of the chosen Himalayan and High Mountain Asia-related regions, the values of the annual balance were grouped by year. Next, a line chart was visualized to check whether the annual balance of glaciers was growing more and more negative each year.

This study helps to compare annual balance and finds whether glacier mass loss has accelerated over the decades. When the mean annual balance become negative between 1990s to 2020s, the mass loss of glaciers has been increasing over the years

3.6.4 Forecasting model

To obtain the forecasting objective, a time-series forecasting approach was employed. Annual balance was the target variable which needed to be forecasted.

The main aim of this forecasting is to predict the future trends in the mass-balance of the glaciers based on past values. This forecasting method can help to identify the glacier mass trend in the future based on the previous mass values of mass-balance.

The forecasting models that can be used in this study are as follows:

- Linear trend model
- ARIMA model
- Exponential smoothing model
- Regression-based forecasting model

A Linear trend model was used as a baseline approach as it helps to show the increasing and decreasing of the annual glacier balance over time. ARIMA and exponential smoothing are also used to predict future trends based on past values.

More advanced machine learning models such as Random Forest or XGBoost can be used in the future if additional attributes such as temperature, precipitation, and snowfall are included. But, the main focus for this dissertation is based on time-series glacier mass-balance trends.

3.6.5 Model Training and Validation

For this analysis, the dataset was divided into sequentially but not randomly. It is more significant because time-series data should be trained using past data and be tested on future data.

To illustrate this, data between 1990 to 2015 can be used for training whereas data between 2016 to 2025 can be used for testing. This allows the model to be learned from the past patterns of mass-balance of glaciers, and then compute how well the model predicts or forecast.

The ordered train-test split is more appropriate than a random train-test split because data on the mass-balance of glaciers is time-dependent. The random splitting may combine the past and future values and provide misleading results.

3.6.6 Model Performance Evaluation

The common regression and forecasting performance measures were used to calculate the forecasting model

These include:

- Mean Absolute Error (MAE
- Root Mean Squared Error (RMSE)
- Coefficient of Determination (R^2)

Mean Absolute Error is the measure that shows the average amount of difference between the actual and the predicted values of annual balance.

Root Mean Squared Error puts more emphasis on the larger errors, and is used when one wants to identify the models that have large prediction errors.

The coefficient of determination shows how much of the variation in annual glacier balance is explained by the model.

One of the characteristics of a good forecasting model is that it has low values of Mean Absolute Error, low values of Root Mean Squared Error and higher values of coefficient of determination. These measures were employed to gauge the effectiveness of the model in predicting the trends in mass-balance of glaciers.

3.7 Data Visualization

Visualization was used for both exploratory data analysis and presenting data in an accessible format. Power Bi was used to create interactive dashboards and visual summaries. To support time-series and forecasting analysis, python visualizations were used.

The main visualization include as follows:

- Glacier count by region
- Total glacier area by region
- Average median elevation by region
- Glacier size-class distribution
- Scatter plots showing area-elevation relationships
- Annual glacier balance trend from 1990 to 2025
- Decade-wise comparison of annual balance
- Regional comparison of annual balance
- Forecast chart of annual balance

These visualization help the interpretation of spatial variability and provide a clear summary of glacier characteristics across High Mountain Asia. They also support the understanding of glacier loss, regional risk and future forecasting.

3.8 Justification for Tool Selection

Power BI was chosen for its excellent data cleaning, interactive filtering, dashboard building and comparative visualization features. It is well suited to the presentation of glacier inventory data and comparing glacier characteristics across regions.

Python was selected because it provides strong libraries such as Pandas, NumPy, Matplotlib, and others for data processing, statistical modelling and visualization which can be useful for time-series analysis and forecasting. Also, it is suitable for analyzing annual glacier mass-balance data.

Excel was used in order to check the data, validate and basic scanning of the datasets.
Therefore, combination of Power Bi, Python, and Excel

3.9 Reliability and Limitations

The use of publically available dataset ensures transparency and reproducibility. The RGI dataset offers a standardized glacier inventory and it is sufficient for spatial inventory analysis. The mass balance dataset provides the historical observations that are required for the time series analysis and forecasting.

However, there are several limitations as RGI data doesn't have a complete annual balance record for glaciers over time, so it is only useful for the glacier characteristics but inappropriate to make forecasts for time-series analysis.

The mass balance dataset contains useful annual records, but lacks some data or the distributions of records among countries and glaciers. So, some regions may have more observations than others, which can affect regional comparison. Furthermore, the division of West, Central, and East Himalaya is a simplified approach and may not capture the physical glacier boundaries

Therefore, forecasting on glacier mass loss is limited due to the climate and environmental factors. So, without complete temperature, precipitation, and snow fall variables, the model mainly forecasts based on annual balance trends. It should be viewed as an approximate analytical value.

3.10 Ethical Considerations

This research is based on the secondary environmental data and published research. No personal or sensitive data were gathered and no human subjects were used. All the data used in the study is referred to properly and employed in academic research only.

All the data and literature sources should be fully cited and attributed. The material presented is solely for educational purposes and is intended to increase knowledge in glacier mass loss and environmental change due to climate change.

3.11 Software and Analytical Tools

The main tools used in this study were Power Bi, Python, and Excel.

Power Bi was used for importing, cleaning and merging the RGI datasets and generating the dashboards and visualization for glacier count, area, elevation, and size-class comparison.

Python was used to process the mass balance dataset, filter the data from 1990 to 2025, compute annual and regional trends, prepare the prediction analysis input, and evaluate prediction performance based on statistical performance indicators.

Excel was used for verifying the column names, missing values and validating the selected output using Excel.

All these tools together supported the overall analytical process of preparation, visualization, exploratory analysis, and forecasting.

3.12 Research Workflow

The overall workflow includes the seven main stages:

- Data collection: Glacier Inventory Data and Mass Balance Data were obtained from secondary sources.
- Data cleaning and preparation: A variety of data inspection such as missing values, inconsistent columns and incorrect datatypes.
- Data integration: RGI regions 13, 14 and 15 were integrated into one single datasets.
- Feature engineering: Several new features were generated such as region, size class, sub-region and decade
- Exploratory visual analysis: Descriptive analysis was applied to summarize glacier count, area and annual balance of the glaciers.
- Regional and time-series analysis: Glacier attributes and mass-balance trends were compared across regions and time frames.
- Forecasting and Evaluation: Annual glacier balance was set as the target variable to forecast and the performance of the model was evaluated using standard metrics

This workflow ensured that the study moved from the raw data collection, preparation and analysis to interpretation and forecasting

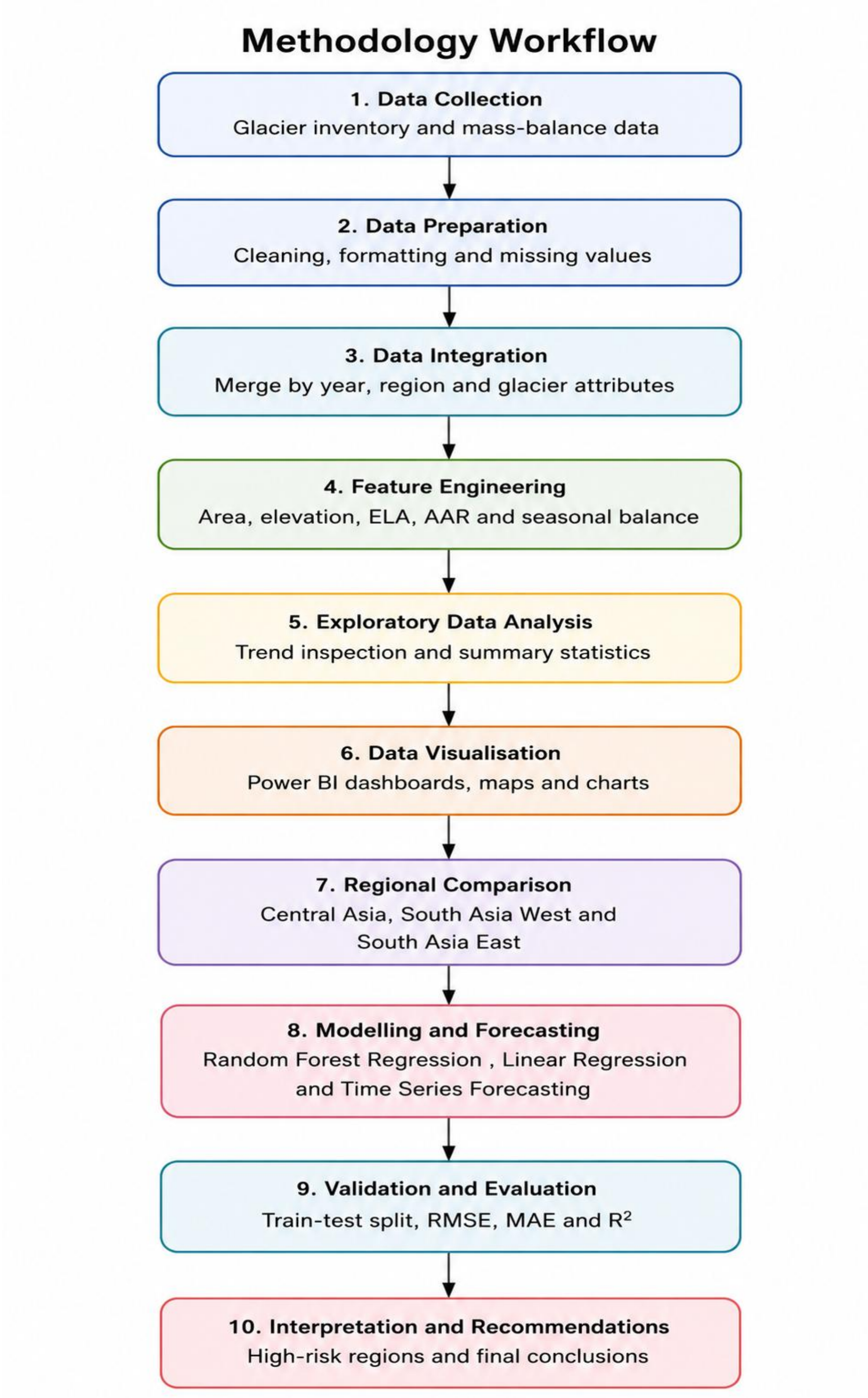

Figure 7: Research Workflow Diagram

3.13 Scope of Analysis

The scope of the study includes both spatial glacier inventory analysis and mass balance time series analysis. The RGI dataset was used to calculate the glacier characteristics across High Mountain Asia, while the mass-balance dataset was used to analyze mass-loss trends from 1990 to 2025.

Glacier area, elevation, regional distribution, and annual glacier mass balance are well addressed within this study. It does not attempt to create a physical glacier simulation, instead, data analytics and forecasting techniques are applied to find regional differences, and to determine trends, regional differences and future mass-loss patterns.

These results from the forecasting should be data driven, pulled from the historical mass-balance observations. For more accurate forecasting in the future, various climate variables such as temperature, precipitation, snowfall, debris cover, and hydrological data are required. However, the methodology gives a suitable analytical answer to the research questions and supports the prediction of future glacier mass loss.

4. RESULT AND ANALYSIS

4.1 Introduction

The aim of this chapter is to present the outcome of the glacier inventory analysis and glacier mass-balance analysis of the Himalayas and wider High Mountain Asia region. The analysis was based on two datasets. Glacier properties such as glacier count, total glacier area, median elevation, and size-class distribution were studied using the Randolph Glacier Inventory data. To calculate the annual trends of the mass-loss of glaciers between 1990 and 2025, glacier mass-balance dataset was studied.

The chapter is separated into two parts. The first part describes the spatial and descriptive analysis across Central Asia, South Asia West and South Asia East. The second part includes the time-series analysis of the annual balance of glaciers, the regional comparison of the glacier mass and the forecasting outcomes. These findings are applied to the research questions in terms of glacier area, elevation patterns, regional differences, acceleration of mass loss and identification of high-risk glacier regions.

4.2 Overview of Glacier Characteristics in High Mountain Asia

A dashboard was created in Power BI to visualize the key characteristics of glaciers in High Mountain Asia. It demonstrates the summary statistics for the total number of glaciers, total glacier area and average median glacier elevation, as well as comparison visuals for the main regions and distribution plots. The dashboard for the regional glacier characteristics in High Mountain Asia are given below:

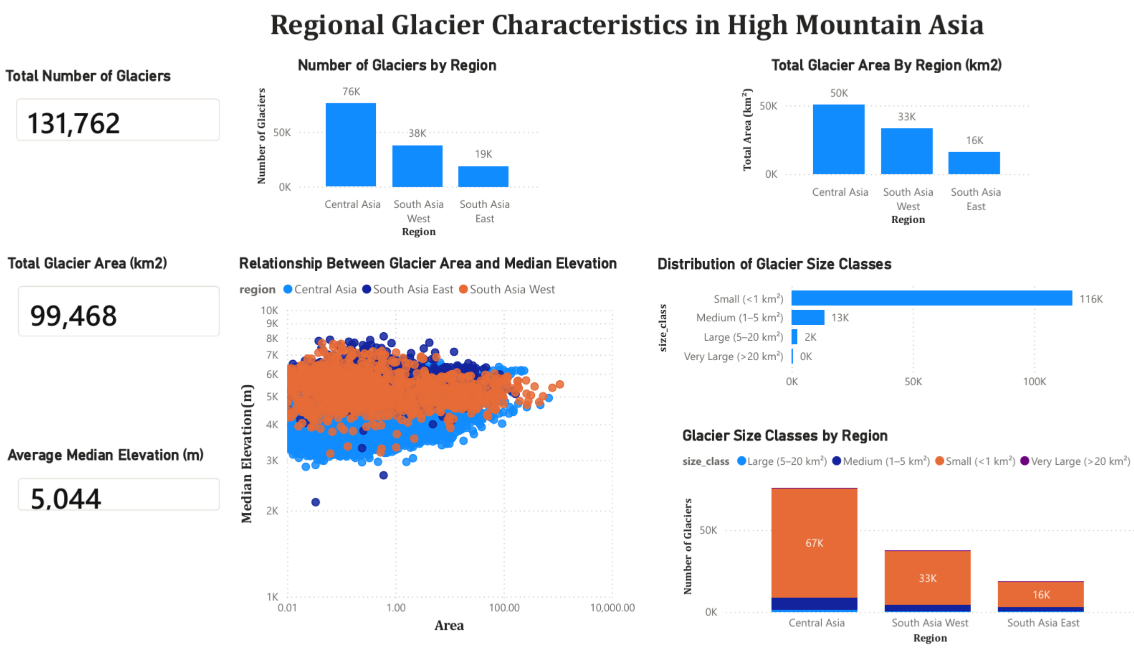

Figure 8: Dashboard for Regional Glacier Characteristics in HMA

The combined RGI dataset shows that the High Mountain Asia such as Central Asia, South Asia West, and South Asia East contain a large number of glacier records across these regions and show the importance of the major glacierized regions outside the polar areas.

For the distribution of glaciers, the analysis is strongly affected by the regional topography. As glaciers are located at the higher altitude, their area, elevation and size_class distribution differs between the regions. These findings illustrate that the glacier systems in High Mountain Asia are spatially complex and cannot be a unique glacier system.

The results provide basic understanding of the physical state of the glaciers before analyzing historical glacier mass-balance and future forecasting.

4.3 Regional Distribution of Glacier Count

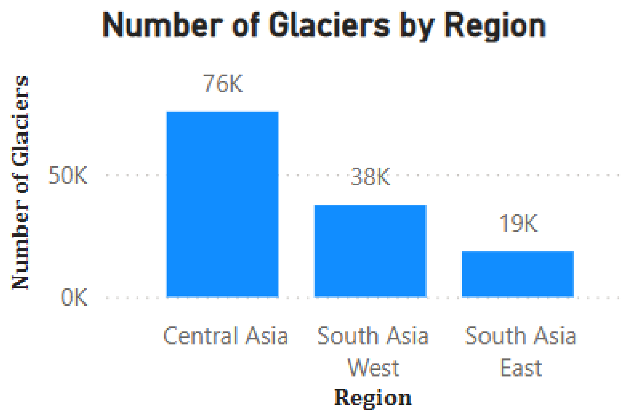

Figure 9: Regional Distribution of glacier count across High Mountain

Figure 9 shows the number of glaciers for each of the three selected regions: Central Asia, South Asia West, and South Asia East. Central Asia has the largest number of glaciers, followed by the South Asia West and South Asia East.

This shows an unequal distribution of glaciers in High Mountain Asia. The huge number of glaciers in Central Asia is due to the broader glacierised terrain, regional climate conditions and topographic environments that promote glacier development. On the other hand, the smaller glacier count in South Asia East indicates a lower number of inventoried glacier units, but does not necessarily suggest lower glacier sensitivity or less glacier importance.

The regional variation in glacier count shows that the spatial distribution of glaciers in High Mountain Asia is uneven. This finding supports the broader view that sub-regional comparison is important in glacier research.

4.4 Regional Distribution of Total Glacier Area

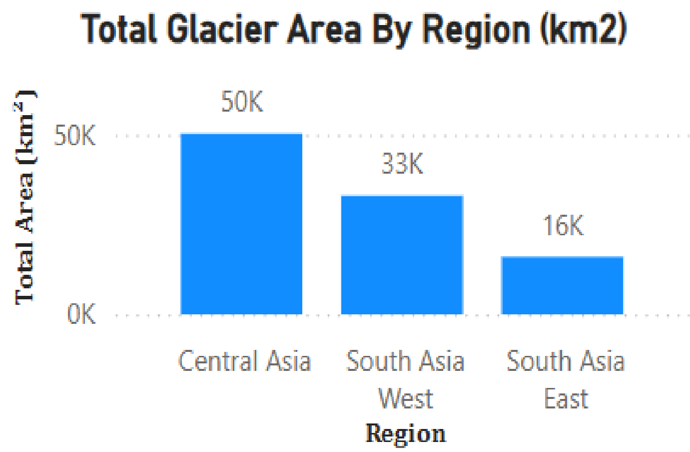

Figure 10: Total glacier area by region in High Mountain Asia

Figure 10 displays the total glacier area for each of the three study region. So, Central Asia has the largest total glacier area, followed by South Asia West and South Asia East having the smallest total glacier area.

The total area of glaciers is an important factor because it represents the spatial events of glacier cover. The number of glaciers is useful, but area is the better measure of the glacier coverage. Therefore, a region might have less number of glaciers, but total glacier area can still be large in regions where individual glaciers are large.

The highest glacier area in Central Asia represent the most extensive glacier coverage within the study dataset. South Asia West contributes a larger glacier area in a major glacierised sub-region within High Mountain Asia. Also. South Asia East includes fewer glaciers and lower total area, still remains as an important glacier region due to its high altitude environments and it's related to Himalayan glacier systems.

Overall, the results suggest that glacier area is inconsistently distributed across High Mountain Asia, with clear concentration in particular regional zones.

4.5 Regional Variation in Glacier Median Elevation

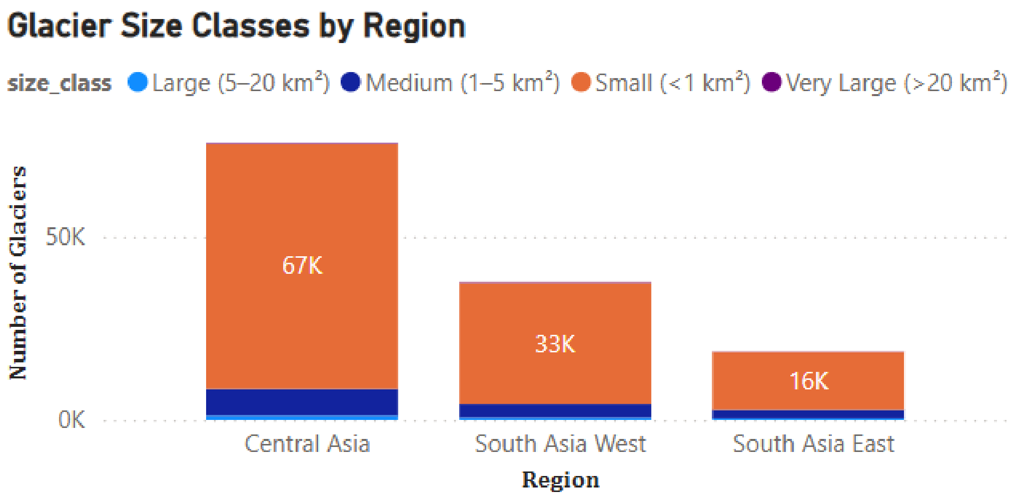

Figure 11: Average median glacier elevation by region

Figure 11 displays the average median elevation of glaciers across the three regions. The results indicate that South East Asia has the highest average median elevation, followed closely by South Asia West, while Central Asia has the lowest average median elevation of the three.

This finding is significant because glacier elevation is closely linked to glacier sensitivity, climatic conditions and accumulation-ablation process. Higher elevation may reflect cooler glacier environments and greater sensitivity to snowfall processes, whereas lower elevations may lead to the melting process under warming conditions.

The high median elevation in South Asia East indicate that glaciers are found at higher elevations in this region due to the steep terrain of the Himalayan and the effect of eastern high-mountain environments. South Asia West also contains high glacier elevations whereas Central Asia has a wider distribution of glaciers due to the low median elevations.

These results show that the glacier elevation patterns differ from glacier count and glacier area counts. So, elevation play an important role in the future glacier mass-loss modelling and forecasting.

4.6 Glacier Size-Class Structure Across High Mountain Asia

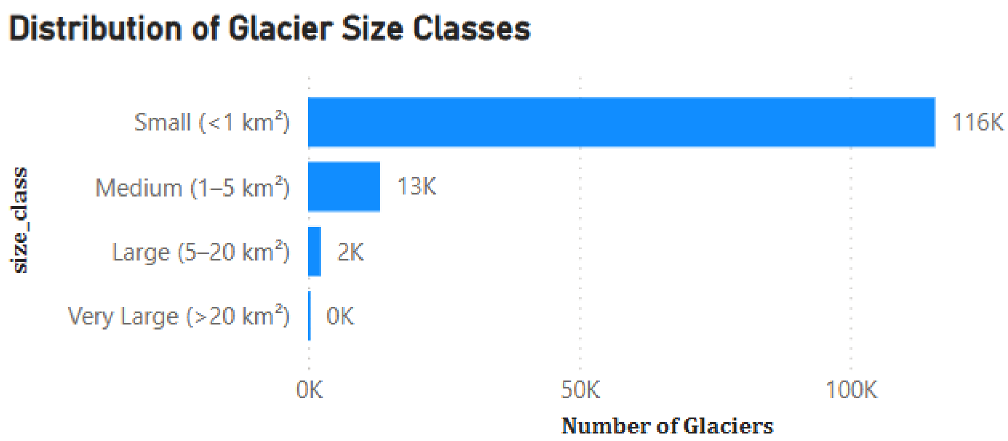

Figure 12: Glacier size-class distribution across High Mountain Asia

Figure 12 displays the distribution of glacier size classes across the merged High Mountain Asia inventory. So, small glaciers dominate the dataset with the highest number, while medium, large and very large glaciers are pretty rare.

From the diagram, the distribution of glaciers in High Mountain Asia is heavily tilted towards the small glacier units. So, they are more sensitive towards climate change because they may respond faster to changes in temperature and precipitation than large glaciers. Therefore, small glaciers may represent a higher threat to future climate change.

The medium, large and very large glaciers are relatively low in number and contribute important ice reserves and they are less in number than small glaciers. This uneven size distribution is common in a glacier inventory and important for understanding sensitivity, glacier stability at a regional scale and future mass-loss potential.

The dominance of small glaciers also indicate that glacier size is valuable in future climate-linked or in prediction models.

4.7 Relationship between Glacier Area and Median Elevation

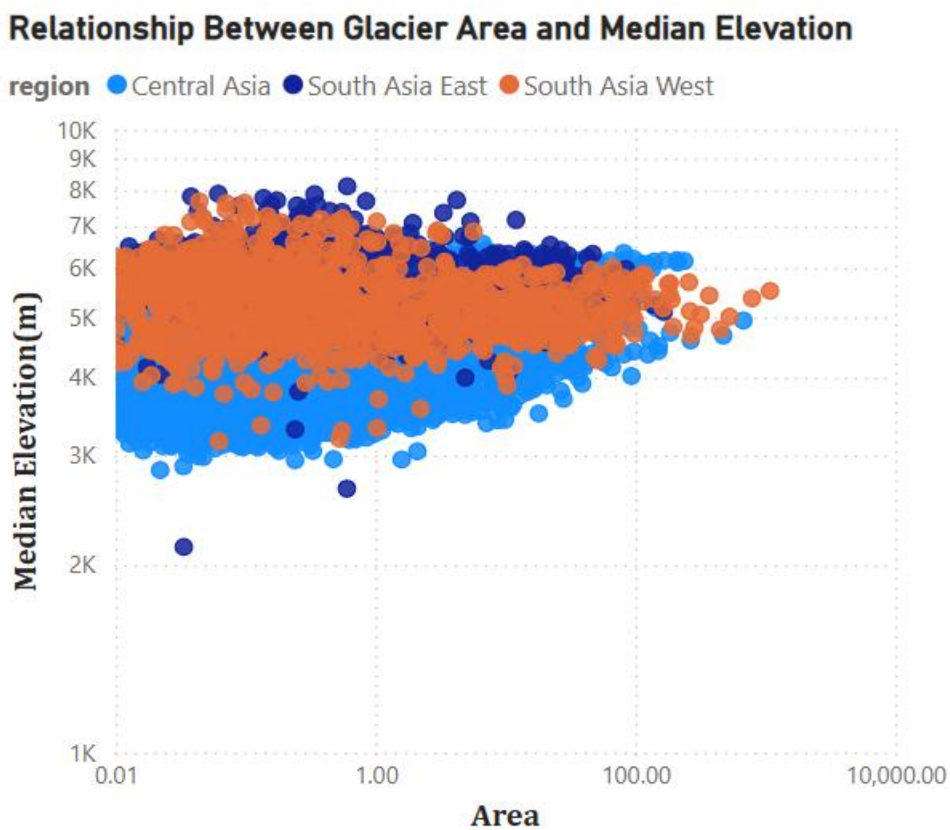

Figure 13: Relationship between glacier area and median elevation

Figure 13 presents the scatterplot showing the relationship between glacier area and median elevation**.** The visual pattern indicates that the majority of the glaciers are situated in the lower range of areas and only few glaciers have reached large sizes. Median elevation differs across this spread, indicating that glacier size and elevation are related but not in a simple linear way.

The scatter plot indicates that a big percentage of the glaciers are small and clustered in a medium range of elevations. The presence of a few very large glaciers is as outliers with far larger glacier areas. This means that there is a much skewed structure of glaciers across the study area.

The findings also make it possible to assume that not all large glaciers are always at the highest altitudes. Rather, glacier elevation seems to be fluctuating with the size of glaciers based on the region and topographical setting. This supports the idea that glacier behaviour is affected by a combination of interacting factors and not by area or elevation alone.

From a logical approach, this scatter plot justifies the use of both the size and elevation of the glaciers as significant explanatory variables in a future glacier mass balance analysis.

4.8 Regional Comparison of Glacier Characteristics

The combined findings indicate that there are significant regional variations in High Mountain Asia. Central Asia has the largest number of glaciers and the greatest total area, indicating that it contains the biggest glacier inventory in the dataset. South Asia West lies between East Asia and Central Asia in terms of the number of glaciers and the area covered by the glaciers, which means that it has a large but decreased total glacier cover. South Asia East consists of less number of glaciers and total glacier area, but it shows the highest average median elevation, highlighting the significant of topographic differences between regions.

These results shows that the glacier systems across High Mountain Asia are not uniformly distributed. Instead, regions exhibit a unique glacier profile with respect to the abundance of glaciers, extent of glaciers and the elevation features. This support that the analysis of glaciers at the regional level is necessary and the change of glaciers must not be viewed as a one-dimensional process across the wider Asian high-mountain system.

The findings also indicate that the different region may differ in glacier vulnerability. Area with smaller glaciers could be more responsive to warming and those with large glaciers areas could provide more important contributions to the overall mass loss. Regions with high median elevation may hold glacier ice at higher altitudes, but they may still remain sensitive to changes in snowfall seasonality and long-term climatic forcing.

Therefore, regional glacier comparison provides an important foundation to determine the risk of future mass-loss of glaciers.

4.9 Time-Series Analysis of Glacier Mass Balance

The time-series of a glacier mass balance was presented based on the mass-balanced point dataset. The balance was the main variable used for the analysis. A negative balance means that the glacier are losing mass and a positive value means they are gaining mass. Firstly, data was filtered and selected from the sub-regions of the Himalayas and observations are available from 2005-2025. The time-series dashboard has been created from the existing data and displayed as given below.

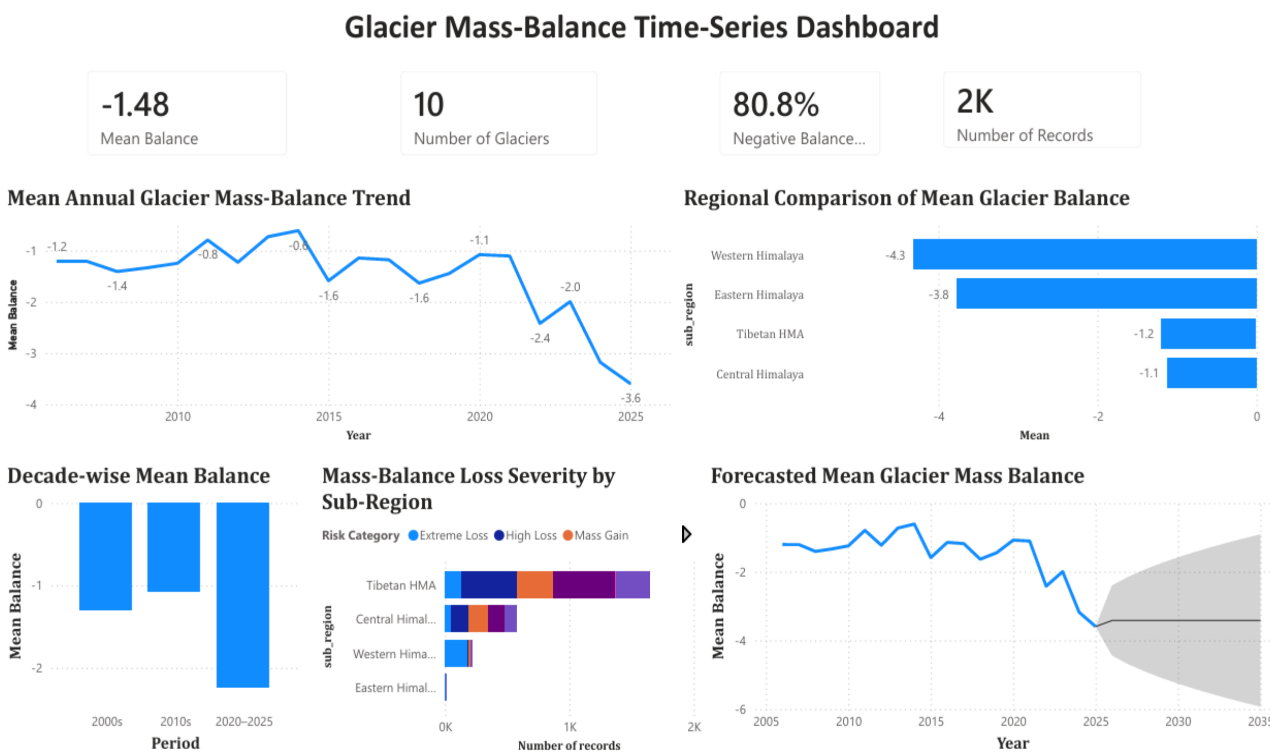

Figure 14: Power BI glacier mass-balance time-series dashboard

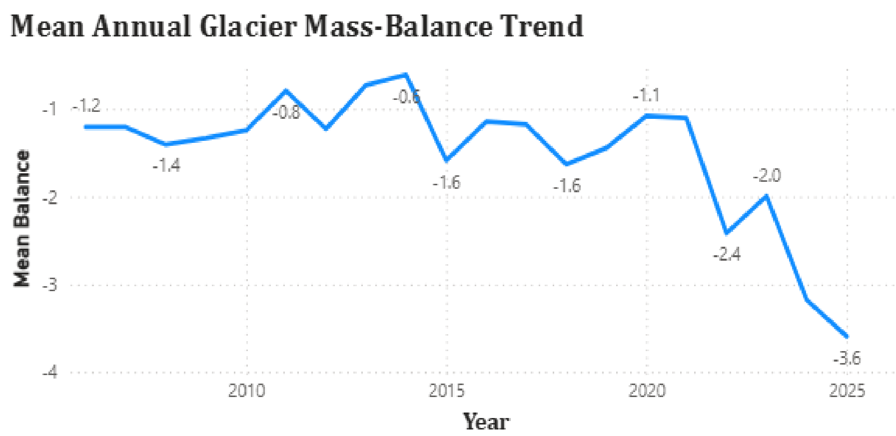

Figure 15: Mean annual glacier mass-balance trend from 2005 to 2025

From above figure, the mean annual glacier mass-balance trend for the study regions. The results illustrate that the glacier mean balance continued to be negative during period of observation**.** This indicate that the selected glaciers experienced continued mass-loss throughout the time. Although, annual values fluctuate from year to year, the overall pattern display constant negative mass balance.

However, the trend turns negative from 2020. The mean balance went from -1.1 in 2020 to -3.6 in 2025 as observed in the dashboard. This concludes that increased glacier mass loss in recent times. Also, the negative values throughout the period indicate that the mass loss events of glaciers are not isolated but is a continuous trend in the sub-regions of the Himalayas.

The dashboard also displays a mean of -1.48, verifying that the mean of the observations chosen is indeed negative. Furthermore, 80.8% of records have negative balance values indicating that the majority of observations are of glacier mass loss, not mass gain. This is a good indication that the filtered data set is dominated by glacier mass loss.

4.10 Regional Comparison of Glacier Mass Balance

The sub_region variable is created from the dataset while comparing the regions of Himalayas. The sub_regions are Western Himalaya, Central Himalaya, Eastern Himalaya, and Tibetan HMA. Those areas were employed to compare glacier mass-balance patterns of the different glacier environments in the Himalayas.

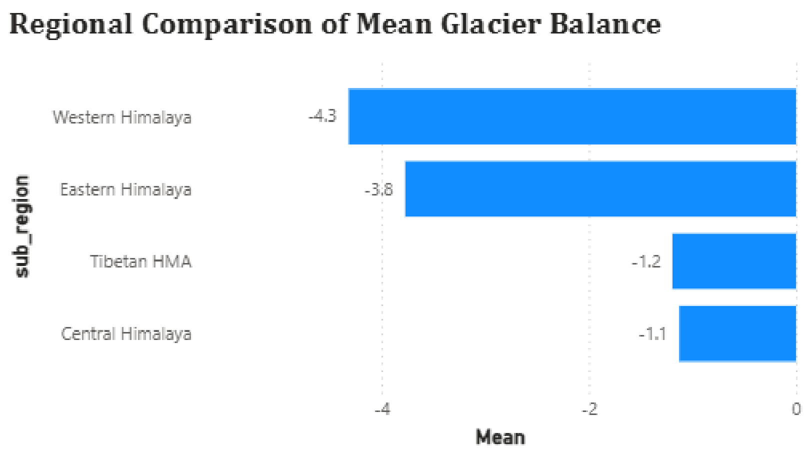

Figure 16: Regional comparison of mean glacier mass balance by sub-region

From figure 16, the mean balance is the lowest for the Western Himalaya at around -4.3, indicating highest glacier mass loss across other regions. The mean balance for the Eastern Himalaya is also -3.8. So, the outcome from glaciers which are in the Western and Eastern are more vulnerable to a strong mass loss.

The Tibetan HMA region has a mean balance of -1.2, while the Central Himalayas has a mean balance of approximately -1.1. It shows that these values are still negative than eastern and western, they still represent overall glacier mass loss.

The results indicate that the glacier mass balance is not consistent throughout the study area. The results help answer the research question on whether melting rates of glaciers differ between the sub-regions of the Himalayas. The results showed that there are some areas where the negative balance is higher than other areas, which points to the need for regional analysis. The number of records varies between sub-regions, though, and care must be taken in interpreting the results.

There is variation in the number of observations across the regions, which may bias the regional averages. The regional comparison should then be regarded as an indicator of spatial variation but it should not be considered as a full physical risk assessment.

4.11 Time-Series Forecasting Results

A forecasting analysis was performed in Power BI with the time series glacier balance. The aim of this forecast is to analyze the possible future loss of the glaciers in the light of the mass-balance history.

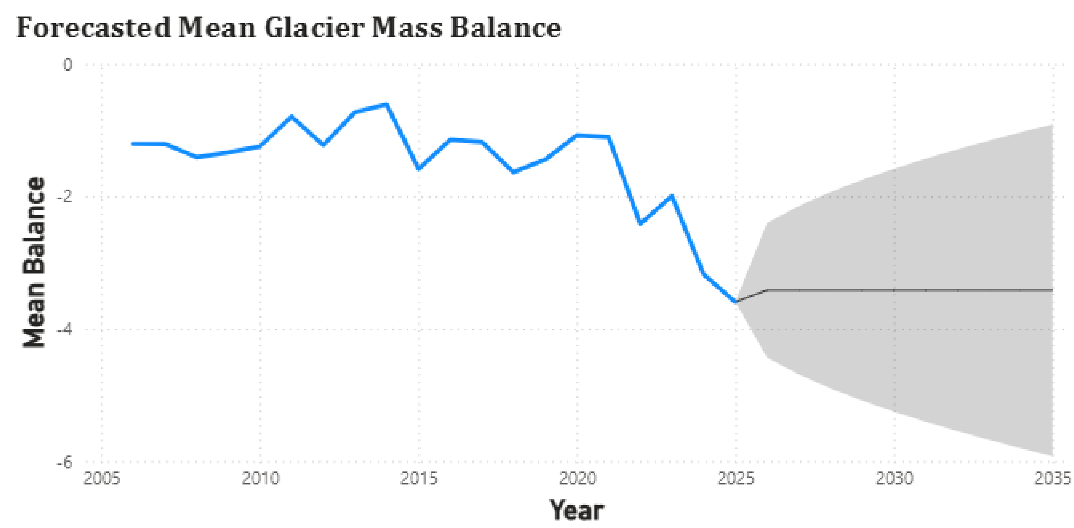

Figure 17: Forecasted mean glacier mass balance

During the time of observation from 2005, mean glacier balance was negative, and increased in negativity since 2020 as shown in the historical line. The Power BI forecast is used to predict future values of the balance based on the previous observed time series with a confidence interval

Glacier balance is expected to be negative in the upcoming years as indicated in the forecast. Therefore, it is expected that if this historic trend is maintained, glacier mass loss will continue. From the above figure, the forecast line is still below zero, so the glacier mass balance is still predicted to below zero, showing glacier loss.

But, the grey confidence interval is wide and shows a lot of uncertainty surrounding the forecast. This uncertainty comes as glacier mass balance is determined by a number of factors such as temperature, precipitation, snowfall, elevation, debris cover and local glacier conditions. These factors are not present in the time-series forecast that is built in Power BI.

Therefore, the forecasting outcome is useful to help indicate the likely direction of future glacier mass balance, but should not be treated as a physical simulation of glacier behaviour.

4.12 Model Performance Comparison

Mean Absolute Error, Root Mean Squared Error and the coefficient of determination was calculated to measure the performance of the machine learning models. So, Linear Regression and Random Forest Regression were used. Linear Regression model was used as a baseline model, while Random Forest Regression was used to capture the non-linear relationships between glacier balance and the independent or predictive variables

The balance was used as target variable for predicting the model representing annual glacier mass-balance value. Negative values of balance shows glacier mass loss, while positive values indicate gain of glacier mass. Therefore, balance was chosen as main output variable because it reflects the change status of annual glacier mass.

The input variables used in this model were year, sub-region, glacier area, equilibrium line altitude (ELA), accumulation-area ratio (AAR), winter_balance and summer_balance. These variables were selected because they provide key information about the temporal, spatial and seasonal nature of glacier mass balance. So, the year variable was used to represent the time trend while the sub-region variable assist to capture regional variation in glacier behavior. Glacier area, ELA, AAR , winter balance, and summer balance have been selected since they are directly related to glacier accumulation and ablation processes.

Pre-processing has been done by removing incomplete records in the target variable and selecting the input variables to build the model. This is needed for a model to maintain accuracy, otherwise due to the missing values or errors in the training process, it might lose accuracy. Sub-region was treated as categorical variable, but then converted to a numerical one for the regression models. This enabled the models to take into account regional variations in the prediction process.

The data was split chronologically due to its time-related nature, instead of randomly. This is more appropriate for time-series and forecasting analysis as the model should be trained on previous records and tested on future records. The test data were records from the years 2021-2025 and the training data were records from the years 2005-2020. This allowed the model to be trained on the historical glacier mass-balance data, and to then test its prediction against more recent data.

From the results, Random Forest Regression performed better than Linear Regression model across all evaluation metrics. Linear Regression produced a Mean Absolute Error of 0.8725, Root Mean Squared Error of 1.0722 and R2 value of 0.6133. This clearly explains that the model explained approximately 61.3 percent of the variation in glacier balance.

Random Forest Regression produced a Mean Absolute Error of 0.5561 and Root Mean Squared Error of 0.7895, which is much lower than the previous model. It achieved a higher R2 value of 0.7903 indicating the model explained about 79 percent of the variation in glacier balance. This concludes that Random Forest Regression gives more exact predictions than Linear Regression.

The model performance of the Random Forest Regression model was better than other classifiers because the glacier mass balance is affected by non-linear relationships between variables such as a year, elevation, location and sub-region. This is expected since glacier mass loss is influenced by multiple interacting physical and climatic conditions rather than by a simple linear trend line.

The table for the comparison between the two models are as given below:

	MODEL	**MAE**	**RMSE**	**R2**
0	Linear Regression	0.872508	1.072167	0.613303
1	Random Forest Regression	0.556138	0.789518	0.790314

Figure 18: Model performance comparison between Linear regression and Random Forest Regression

Based on these results, Random Forest Regression was chosen as the better performing model for predicting glacier mass balance than other models. However, the model results should be taken carefully because the dataset doesn't contain all climate variables such as temperature, precipitation and snowfall which can affect the glacier mass balance.

4.13 Forecasting Results and Risk Interpretation

The forecasting and regional results give valuable insights into potential glacier mass-loss risk in the future. The mean balance for all glaciers is -1.48, which means that they are experiencing a net loss of mass. The overall negative balance percentage is around 80.0% shows that most observations indicate glacier loss.

In recent years, from 2020 in general, the time-series trend indicates that the glacier balance has become more negative. The mass loss of glaciers has risen in the latest phase of the observed period. According to the forecast chart, glacier balance is expected to be negative in future years, manifesting that glaciers are likely to continue losing mass in the future if the same pattern continues.

The regional comparison is useful for detecting sub-regions that have more mass-loss characteristics. The most negative mean balance is found in the Western Himalaya followed by Eastern Himalaya. Also, the negative mean balance was also found on Tibetans HMA and Central Himalayas, suggesting glacier mass loss in all sub-regions. However, the mass loss intensity is lower than Western and Eastern ones.

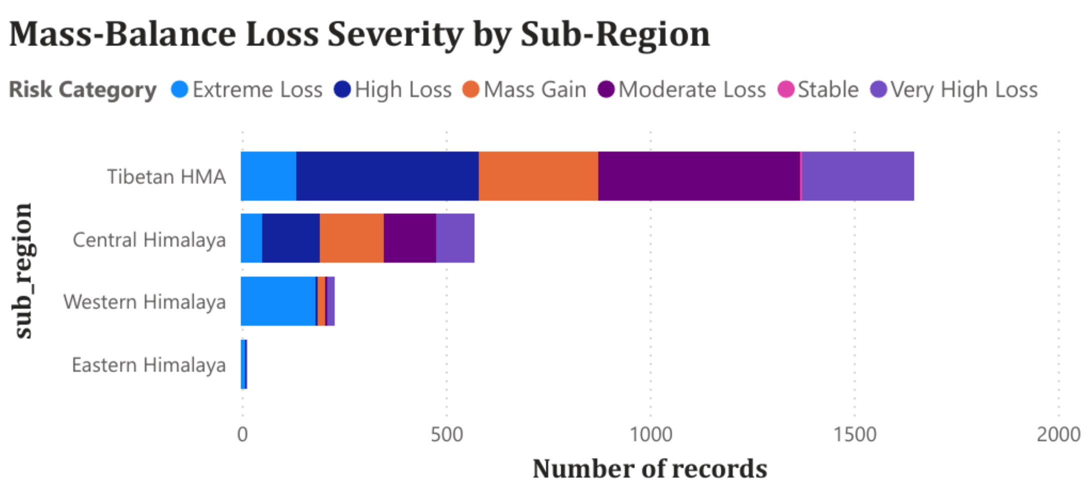

Figure 19: Mass-balance loss severity by sub-region

This interpretation is further understood from the loss-severity chart that illustrates the distribution of records in the various severity categories of mass-balance. The proportion of high loss or extreme loss records in these regions can be interpreted as being more vulnerable to accelerated glacier loss. This classification is based solely on mass-balance values, and should be viewed as an analytical hazard assessment indicator but not as a full hazard assessment.

Overall, the results suggest that glacier mass-loss risk is spatially uneven. Western and Eastern Himalaya have higher negative values than Central Himalaya and Tibetan HMA.

These findings support the idea that the glacier loss in the Himalayas differs by region and the monitoring should be more focused on regions with high negative mass balance patterns in the future.

4.14 Relevance to the Research Questions

The conclusion of this chapter is directly related to the research questions of the dissertation.

The first research question is related to the change in glacier area and elevation in the Himalaya in the last decade. The previous inventory study provides a baseline understanding of glacier area, elevation and size-class distribution across High Mountain Asia. A mass-balance analysis doesn't measure area and elevation change in each year, but it supports the wider investigation of glacier change by showing an indication of change over time. This question is half addressed through glacier inventory analysis and supported by mass-balance trend analysis.

The second research question was whether there was any significant difference in glacier melting rates across different sub-regions of the Himalayas. The comparison of the regions reveals that the mean glacier balance differs across the sub-regions. So, Western and Eastern Himalayas are found to be more negative than the Central and Tibetan parts of the HMA. This confirms that glacier loss balance is not uniform across those sub-regions.

The third research question is about whether the rate of glacier loss in the Himalayas is accelerating over time. The time series chart exhibits that mean balance remained negative throughout the observed period and become more negative after 2020. The decade-wise chart shows that the 2020-2025 period is the most negative in terms of mean balance. This shows that glacier mass loss has increased over the last few years.

The fourth research question was which regions are at high risk of accelerated glacier loss. From the observation, Western and Eastern Himalaya have the lowest negative mean balance values. These regions are considered as higher mass severity in the dataset. However, this interpretation is based on mass-balance records and should be considered as an analytical indicator rather than a complete regional hazard assessment.

Overall, these results address the research questions by using inventory analysis, time-series analysis, regional comparison and forecasting. It summarizes that glacier mass loss is ongoing, differs regionally and is likely to continue if this trend remains unchanged.

4.15 Chapter Summary

This section provides the outcome of glacier inventory analysis, mass-balance time series analysis, regional comparison and forecasting. From the glacier inventory analysis, Central Asia West and South Asia East have different glacier characteristics in terms of glacier count, total glacier area, median elevation and size-class compared to Central Asia East.

From the mass-balance dashboard, the overall negative mass balance for the selected sub-regions of the Himalayas and High Mountain Asia has a mean value of -1.48. It also demonstrates that around 80 percent of records are negative which indicates glacier mass loss in the majority of the observations.

During time series analysis, the mean annual glacier balance has a negative balance from 2005 to 2025 and becomes more negative after 2020. The decade wise comparison shows that the 2020-2025 year had the strongest negative mean balance indicating that glacier mass loss has boosted in recent years.

The regional comparison showed that Western Himalaya and Eastern Himalaya had the most negative balance values, indicating the severity of mass loss was greater in these regions. Also in the Central Himalayas and Tibetan HMA have negative balance values with less severe average loss.

Analysis of the forecast suggests that glacier mass balance is likely to be negative in future years. And, the wide confidence interval in the forecast indicates high uncertainty in the next period. So, this is not an accurate prediction, but rather is an initial evaluation.

In summary, the findings signify that the glacier mass loss in those sub-regions of the High Mountain Asia and the Himalayas is ongoing, unevenly distributed and expected to become a significant issue in the near future.

5. DISCUSSION, CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter addresses the key findings of the dissertation in relation to the research aim and the research questions. The main aim of the study was to analyze and forecast glacier mass loss in the Himalayas and High Mountain Asia based on glacier inventory data, observations of glacier mass balance, Power Bi visualization and time-series forecasting. This section explains the results given from Chapter 4 which is related to glacier mass-loss analysis, identifies the limitations of the study and provides the recommendation for future studies. The discussion is carried around the key main themes such as glacier inventory characteristics, regional variation, time-series mass-balance trends, forecasting results and high-risk glacier regions.

5.2 Discussion of Glacier Inventory Findings

These findings suggest that these glacier systems are not uniformly distributed. The region consists of various environments with different topographic and climatic conditions. So, glacier change should not be analyzed as a uniform across the whole region. Therefore, regional comparison is needed to know the characteristics of the glaciers and other necessary differences in glacier response.

In terms of glacier size-class, small glaciers are high in number in an inventory and it is important because they are more sensitive to climate change due to the low volume of ice and may respond faster to changes in temperature and precipitation. But, large glaciers are fewer in number, they are also valuable because they contain significant ice reserves and can contribute huge to total mass loss.

From the scatterplot between glacier area and median elevation, it shows the complicated and non-linear relationship between them. There are a large number of small glaciers and only a few large ones. This supports the idea that glacier behavior is influenced by glacier size, elevation, region and local climate conditions.

5.3 Discussion of Time-Series Mass-Balance Findings

The time-series analysis of glacier mass balance across the Himalayan and High Mountain Asia regions had negative mass balance. The overall mean balance was -1.48, which means the observations on the selected glaciers were mostly dominated by mass loss. About 80 percent of records in the dashboard show negative balance values, meaning that most observations represented glacier mass loss rather than mass gain.

The annual trend chart demonstrates that glacier mass balance has become negative throughout the observed period. The filtered dataset was focused on the period from 2005 to 2025 because all the usable records of the Himalayan and High Mountain Asia are available mostly from 2005 onwards.

The results indicated that glacier balance has become more negative after 2020. It shows that the rate of glacier mass loss has risen over the last few years. In case of decade-wise comparison, the mean balance was found to be most negative in the 2020-2025 period. This also addresses the research question, asking whether glacier mass loss is accelerating over time.

However, the trend should be kept in mind as glacier mass balance is affected by many climate and physical variables such as temperature, debris cover, snowfall, precipitation, elevation, slope and local glacier conditions. Thus the shown negative trend indicates an increase in mass loss, but not fully explained the all causes of glacier change.

5.4 Discussion of Regional Variation

Glacier mass balance is found to be different among the sub-regions in terms of regional comparison. Western Himalaya and Eastern Himalaya demonstrate the most negative mean balance in the Power BI dashboard, followed by Eastern Himalaya, Central Himalaya and Tibetan HMA with negative values, but with lower mean values of mass loss.

This finding is consistent with previous results that glacier mass loss is not the same throughout the study area. Mass loss seems to be greater in some sub-regions than in others. This helps in answering the research question that is whether there is significant difference in the glacier melting rates between the different parts of the Himalayas.

Western Himalaya had the highest negative mean balance in the dashboard. This indicates the more serious the severity of mass loss in the available data set. The eastern Himalaya region also had excellent negative balance, suggesting glacier loss is also serious in the eastern portion of the Himalayan system. Less negative mean values, but overall glacier mass loss was observed in Tibetan HMA and Central Himalaya.

The regional comparison also identified a key data constraint. There were more observations in some sub-regions than others. For instance, Hindu Kush was excluded from the last dashboard due to insufficient data to analyze it. This illustrates that the results at a regional level need to be interpreted with care, particularly if the number of observations is low.

5.5 Discussion of Forecasting Results

Forecasting was done using the time-series forecasting function in Power BI and machine learning regression models. Using PowerBI, a visualization time series forecast was produced and Linear Regression and Random Forest Regression were used for glacier mass balance prediction based on year, elevation, latitude, longitude and sub-region.

Linear Regression and Random Forest Regression are the two machine learning models with alongside the Power BI forecast. Random Forest Regression was found to outperform the Linear Regression model in the model comparison. The outcomes of the two models were compared and compared to the standard deviation of the data. Random Forest was able to produce a lower MAE of 0.5561 and lower RMSE of 0.7895 whereas Linear Regression was able to produce an MAE of 0.8725 and RMSE of 1.0722. Both models were compared with the standard deviation of the data which was 0.7172. Random Forest also yielded a higher R2 of 0.7903 which explained about 79.0% of the variance in glacier balance. This indicates that there may be non-linear relationships between year, elevation, location, and sub-region on glacier mass balance. Therefore, Random Forest was more suitable than Linear Regression for glacier mass-balance prediction in this study.

The forecast is valuable since it is related to the focus of the dissertation, which is to predict glacier mass loss. It is an indicator of the historical negative balance pattern, which is likely to continue. The forecast was forecasted, however, with a large range of confidence intervals, which means there is a lot of uncertainty in future estimates.

The uncertainty is normal as Power BI forecasting relies on the patterns in the historical data, but not fully on all physical and climatic variables affecting glacier behaviour. Time is not the only factor that affects glacier mass loss. Also affected by global warming temperature, snowfall patterns, precipitation, debris cover and topography of glaciers. So, the prediction should be expected as an exploratory estimate based on the nature of the trend rather than as a physical prediction.

More advanced forecasting methods can be developed using ARIMA, exponential soothing, Gradient Boosting or XG Boost and by including climate variables like temperature, precipitation, and snowfall.

Forecasting was done using the time-series forecasting function in Power BI and machine learning regression models. Using PowerBI, a visualization time series forecast was produced and Linear Regression and Random Forest Regression were used for glacier mass balance prediction based on year, elevation, latitude, longitude and sub-region.

5.6 Discussion in Relation to Research Questions

The first research question was about how glacier area and elevation have changed in the Himalayas over the past decades. The glacier inventory analysis provided a baseline understanding of glacier area and elevation across High Mountain Asia. The analysis showed that glacier area and median elevation vary across regions. However, the RGI dataset is mainly an inventory dataset and does not fully provide repeated annual glacier area and elevation changes from 1990 to 2025. Therefore, this research question was partly answered through spatial inventory analysis, but complete area and elevation change analysis would require multi-year satellite-derived glacier area and elevation datasets.

The second research question asked whether glacier melting rates vary significantly across different sub-regions of the Himalayas. This question was answered through the regional mass-balance comparison. The dashboard showed that Western Himalaya and Eastern Himalaya had more negative mean balance values than Central Himalaya and Tibetan HMA. This indicates that glacier mass loss varies across sub-regions.

The third research question asked whether the rate of glacier mass loss in the Himalayas is accelerating over time. The time-series trend and decade-wise comparison showed that glacier balance became more negative after 2020. The 2020–2025 period showed the strongest negative mean balance. This suggests that glacier mass loss has intensified in recent years.

The fourth research question asked which regions are most at risk of accelerated glacier loss. Based on the available dataset, Western Himalaya and Eastern Himalaya showed the strongest negative mean balance values and can be interpreted as higher mass-loss severity regions. However, this result should be considered as an analytical indicator rather than a complete hazard-risk assessment, because glacier risk also depends on other factors such as glacier lake development, downstream exposure and climate projections.

5.7 Limitations of the Study

There are certain limitations to this study. The RGI dataset is for the most part a glacier inventory dataset. It does offer information that is helpful for glacier area, glacier elevation, glacier location and glacier physical characteristics, however, it is not the complete annual time-series data of glacier area and glacier elevation change. Therefore, the inventory analysis is useful for spatial comparison, but less useful to measure detailed historical change.

Second, the mass-balance data was not equally distributed over the regions and the years. The data set was filtered by selected sub-regions of the Himalayan and High Mountain Asia regions, usable observations were predominantly available from 2005 onwards. This is because the entire proposed period (1990-2025) has not been able to be fully analyzed with the filtered dataset.

Third, there were differences in record numbers between sub-regions. More observations were made in some areas than others and in some areas very few observations were recorded. This will impact the comparison reliability across the region. In areas with low recording density, caution should be exercised in interpretation.

Fourth, the sub-region classification was done using country codes. This is a simplified approach and is not exactly congruent with the physical limits of the Western, Central and Eastern Himalayas. Glaciers are not only counted in the Himalayas but also those parts of glaciers in the larger system of High Mountain Asia.

Fifth, the forecasting model was built with the built-in forecasting function of Power BI. This is helpful for visual trend forecasting, but not all physical and climate variables influencing glacier behaviour are included. The forecast is therefore considered as a "tentative" forecast, not as a physical forecast.

In sixth, using the variables such as year, elevation, location and sub-region, from the dataset, machine learning models were developed. Although, the models doesn’t have full climate variable such as temperature, precipitation and snowfall, it limits the physical interpretations of the model results as glacier mass balance is strongly influenced by the climatic conditions.

5.8 Recommendations for Future Research

Future studies using more complete multi-year glacier datasets should be used to quantify changes in glacier area and elevation between 1990 and 2025. Glacier area data derived from satellites as well as digital elevation model data would provide more direct answers to the research question of area and elevation change.

Climate variables like temperature, precipitation and snowfall should be added to future studies. These variables would be useful in understanding how glaciers change in mass balance over time, and why some parts of a glacier lose mass more than others.

Other, more sophisticated, weather prediction models should also be used. Various models, including ARIMA, exponential smoothing, and Random Forest, Gradient Boosting and XGBoost were considered and compared to determine which model performed best in predicting glacier mass balance. The Mean Absolute Error, Root Mean Squared Error and R^2 could be used to assess model performance.

There is also a need for further research to enhance regional classification. Glaciers could be classified by exact mountain ranges rather than country code, and/or glacier basin boundaries. This would give a better comparison of the situation in the Western, Central and Eastern Himalayan regions.

In conclusion, glacier lake and glacier hazard data could be included in future studies. This will aid to link the glacier mass loss with downstream risks such as glacial lake outburst flood and vulnerability of water resources.

5.9 Conclusion

This dissertation analyzed glacier mass loss in the Himalayas and wider High Mountain Asia region using glacier inventory data, glacier mass-balance observations, Power BI visualization and time-series forecasting. It revealed that glacier numbers, glacier areas and the medians of glacier elevations and size classes in High Mountain Asia are highly variable.

From the mass-balance analysis, it was concluded that the glacier balance in most of the selected sub-regions was largely negative. The mean overall balance, displayed on the dashboard, was equal to -1.48, and the balance percentage was negative for 80.8% of the observations meaning most observations were glacier mass loss. The trend in the time series indicated a more negative glacier balance following 2020, indicating a more pronounced glacier mass loss in recent years.

The regional comparison revealed that the mean balance values in the filtered dataset were highest in the negative in Western Himalaya and Eastern Himalaya. This implies severity of mass losses in these regions might be more compared to Central Himalaya and Tibetan HMA. The results should, however, be treated with caution as the data is not available for every region.

The analysis of the forecasting indicated a possibility of continued negative glacier mass-balance in future years. This means that glacier mass loss will probably continue for the foreseeable future under conditions of continued historical glacier mass loss rates. The forecast was, however, accompanied by high uncertainty and is not a prediction, but an estimate based on trends.

The machine learning results indicate that Random forests regression perform better than Linear Regression while predicting glacier balance. As, Random Forest had an R square of 0.7903 while Linear Regression has an R square of 0.6133. Therefore, Random forest can describe a bigger part of the glacier balance variability and suitable for modelling non-linear relationship between glacier balance and the selected explanatory variables.

Overall, the dissertation indicates that glacier mass loss is persistent in the selected sub-regions of the Himalayan region and High Mountain Asia and is expected to be an environmental problem across the region. The study illustrates the application of data analytics, visualization in Power BI and time series forecasting to facilitate monitoring and analysis of climate change in glaciers.

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APPENDICES

Appendix A: Power BI Dashboard Access

For regional glacier characteristics in HMA,

https://app.powerbi.com/links/dYq_txFgbp?ctid=3d1cee9c-8bf0-4375-b5b9-5c0315d1187e&pbi_source=linkShare

For mass balance time-series dashboard,

https://app.powerbi.com/links/vBSvmXcILa?ctid=3d1cee9c-8bf0-4375-b5b9-5c0315d1187e&pbi_source=linkShare

Appendix B: Python Script from Google Collab

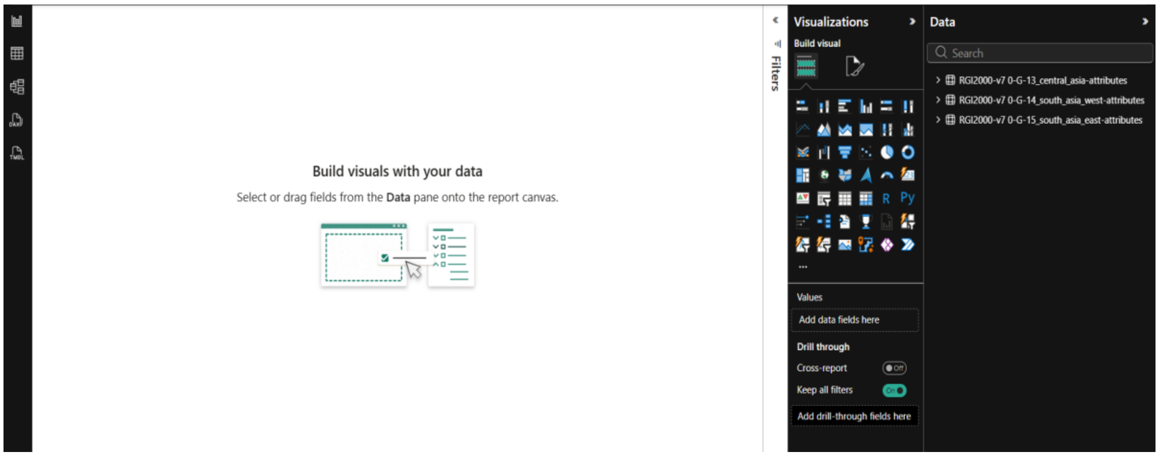
```python
#Importing libraries
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor

# Load mass_balance dataset
df = pd.read_csv("/mass_balance_point.csv")

# Filter years from 2005 to 2025
df = df[(df["year"] >= 2005) & (df["year"] <= 2025)]

# Keep useful countries / regions
df = df[df["country"].isin(["IN", "NP", "BT", "CN"])]

# Remove missing target values
df = df.dropna(subset=["balance"])

# Create sub-region
def region(country):
    if country == "IN":
        return "Western Himalaya"
    elif country == "NP":
        return "Central Himalaya"
    elif country == "BT":
        return "Eastern Himalaya"
    elif country == "CN":
        return "Tibetan HMA"
    else:
        return "Other"
```

```
df["sub_region"] = df["country"].apply(region)

# Select features and target
features = ["year", "latitude", "longitude", "elevation", "sub_region"]
target = "balance"

# Remove rows with missing feature values
df_model = df.dropna(subset=features + [target])

X = df_model[features]
y = df_model[target]

# Split data into training and testing set
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Numeric and categorical columns
numeric_features = ["year", "latitude", "longitude", "elevation"]
categorical_features = ["sub_region"]

preprocessor = ColumnTransformer(
    transformers=[
        ("cat", OneHotEncoder(handle_unknown="ignore"), categorical_features)
    ],
    remainder="passthrough"
)

# Using Linear Regression and Random Forest Regression Model
models = {
    "Linear Regression": LinearRegression(),
    "Random Forest Regression": RandomForestRegressor(
        n_estimators=200,
        random_state=42
    )
}
```

```python
# Train and evaluate the mean absolute error, root mean square error, and r-square
results = []

for name, model in models.items():
    pipeline = Pipeline(
        steps=[
            ("preprocessor", preprocessor),
            ("model", model)
        ]
    )

    pipeline.fit(X_train, y_train)
    predictions = pipeline.predict(X_test)

    mae = mean_absolute_error(y_test, predictions)
    rmse = np.sqrt(mean_squared_error(y_test, predictions))
    r2 = r2_score(y_test, predictions)

    results.append([name, mae, rmse, r2])

results_df = pd.DataFrame(
    results,
    columns=["Model", "MAE", "RMSE", "R2"]
)

print(results_df)
```

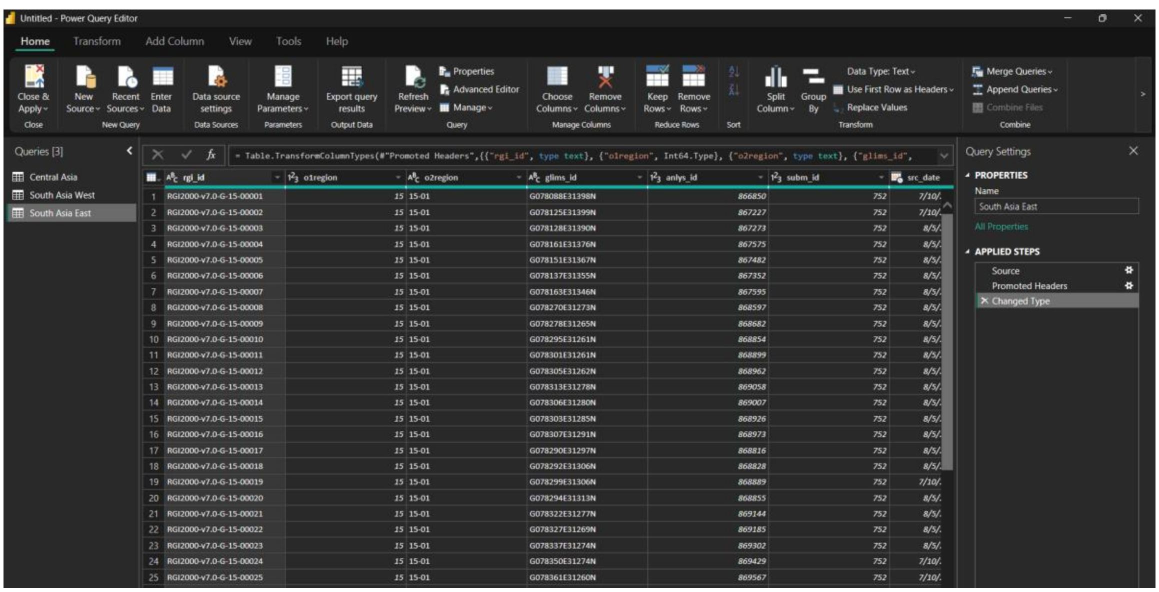
Appendix C: Data cleaning and Preparation in Power Bi

- ## Importing RGI datasets into Power BI

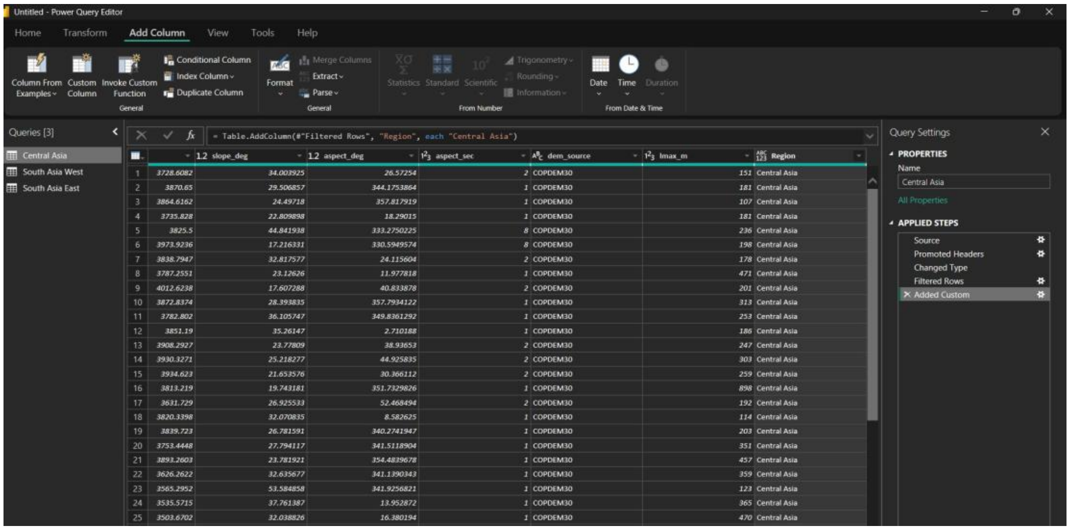


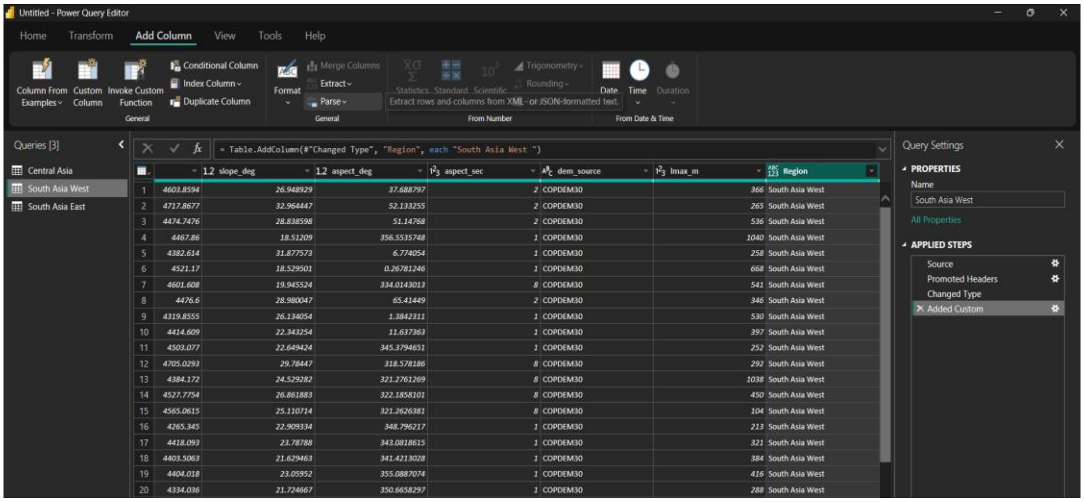
This is the first screen shot of the regional datasets from RGI that were initially imported into PowerBI for further cleaning and analysis and also the glacier mass-balance dataset.

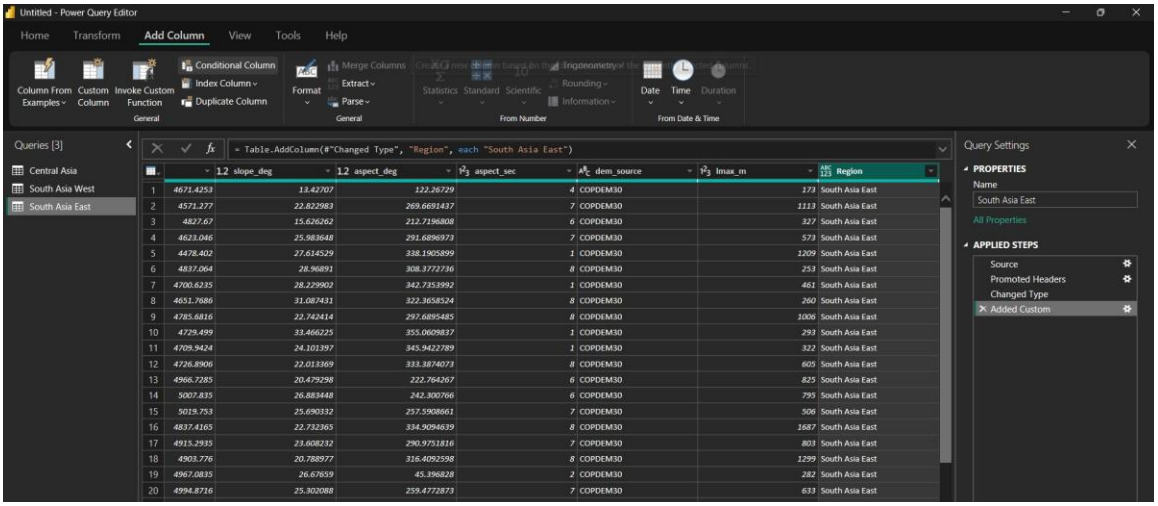
- ## Using Power Query Editor For Data Cleaning



- **Adding region labels for Central Asia, South Asia West and South Asia East**

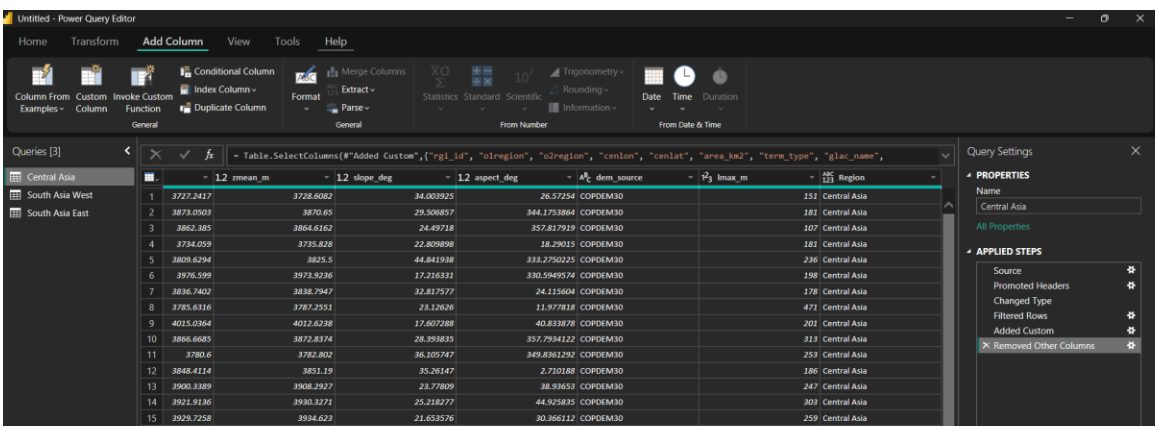


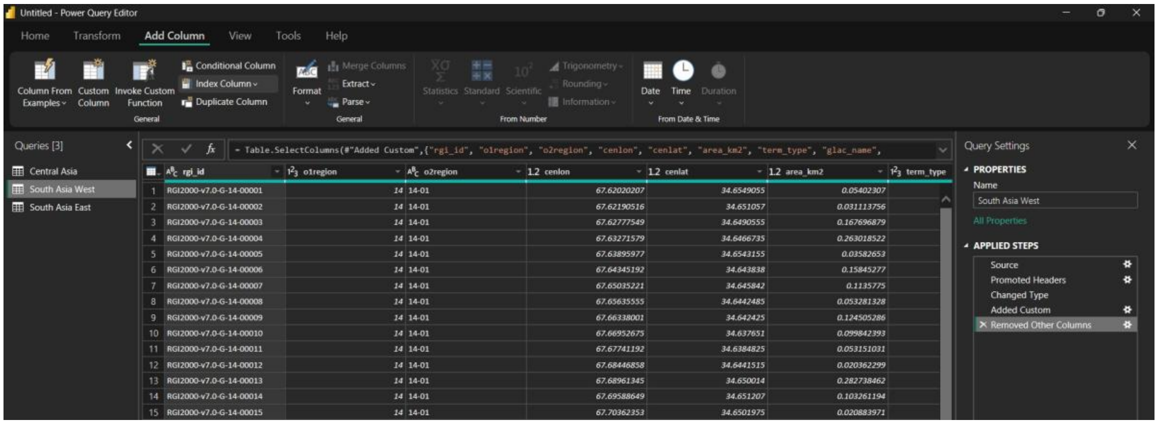




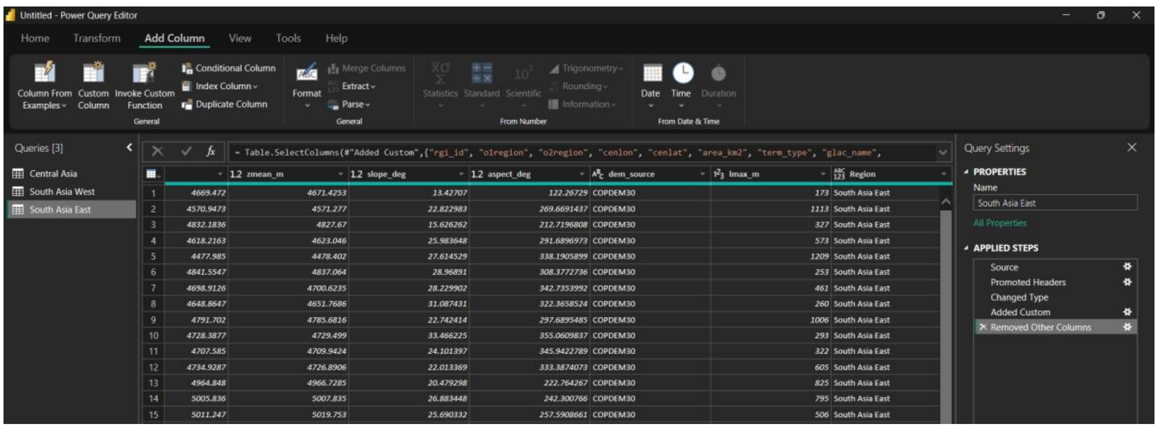
Before merging the datasets, a new Region column was added to indicate each RGI dataset as Central Asia, South Asia West or South Asia East.

- **Removing unnecessary columns from three regions**





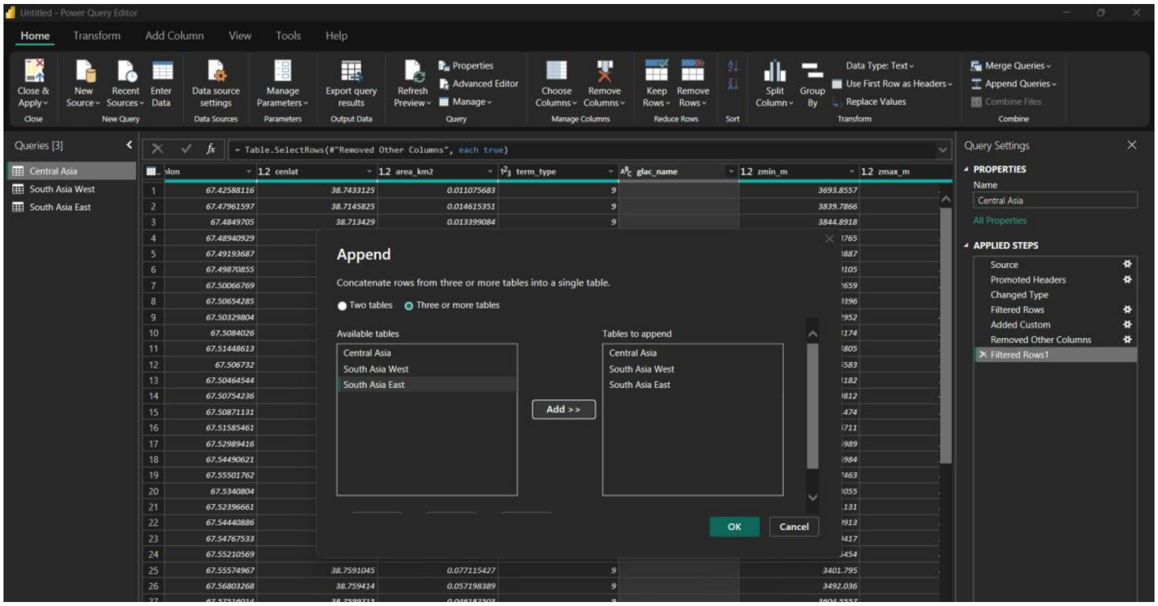
	rgi_id	o1region	o2region	cenlon	cenlat	area_km2	term_type
1	RGI2000-v7.0-G-14-00001	14	14-01	67.62020207	34.6549055	0.05402307	
2	RGI2000-v7.0-G-14-00002	14	14-01	67.62190516	34.651057	0.031113756	
3	RGI2000-v7.0-G-14-00003	14	14-01	67.62777549	34.6490555	0.167696879	
4	RGI2000-v7.0-G-14-00004	14	14-01	67.63271579	34.6466735	0.263018522	
5	RGI2000-v7.0-G-14-00005	14	14-01	67.63895977	34.6543155	0.03582653	
6	RGI2000-v7.0-G-14-00006	14	14-01	67.64345192	34.643838	0.15845277	
7	RGI2000-v7.0-G-14-00007	14	14-01	67.65035221	34.645842	0.1135775	
8	RGI2000-v7.0-G-14-00008	14	14-01	67.65635555	34.6442485	0.053281328	
9	RGI2000-v7.0-G-14-00009	14	14-01	67.66338001	34.642425	0.124505286	
10	RGI2000-v7.0-G-14-00010	14	14-01	67.66952675	34.637651	0.099842393	
11	RGI2000-v7.0-G-14-00011	14	14-01	67.67741192	34.6384825	0.053151031	
12	RGI2000-v7.0-G-14-00012	14	14-01	67.68446858	34.6441515	0.020362299	
13	RGI2000-v7.0-G-14-00013	14	14-01	67.68961345	34.650014	0.282738462	
14	RGI2000-v7.0-G-14-00014	14	14-01	67.69588649	34.651207	0.103261194	
15	RGI2000-v7.0-G-14-00015	14	14-01	67.70362353	34.6501975	0.020883971	



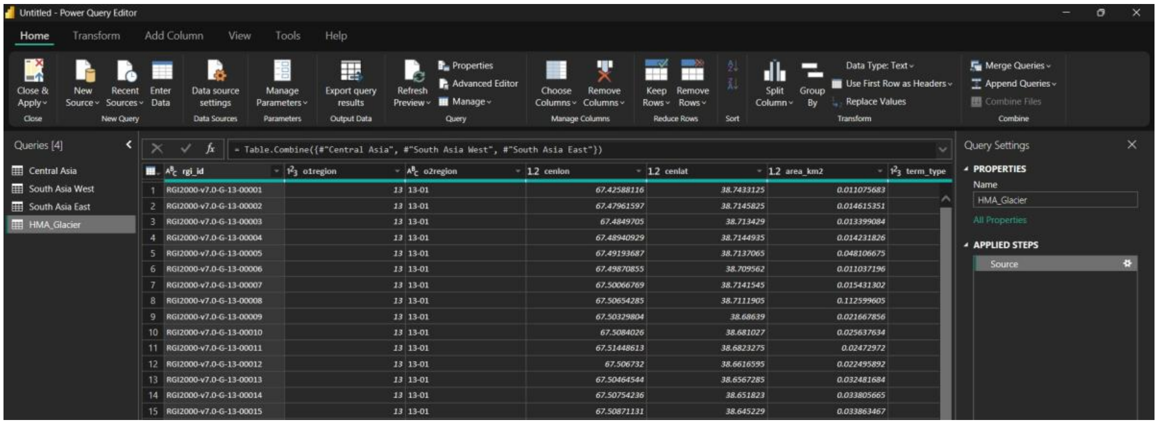
		zmean_m	slope_deg	aspect_deg	dem_source	lmax_m	Region
1	4669.472	4671.4253	13.42707	122.26729	COPDEM30	173	South Asia East
2	4570.9473	4571.277	22.822983	269.6691437	COPDEM30	1113	South Asia East
3	4832.1836	4827.67	15.626262	212.7196808	COPDEM30	327	South Asia East
4	4618.2163	4623.046	25.983648	291.6896973	COPDEM30	573	South Asia East
5	4477.985	4478.402	27.614529	338.1905899	COPDEM30	1209	South Asia East
6	4841.5547	4837.064	28.96891	308.3772736	COPDEM30	253	South Asia East
7	4698.9126	4700.6235	28.229902	342.7353992	COPDEM30	461	South Asia East
8	4648.8647	4651.7686	31.087431	322.3658524	COPDEM30	260	South Asia East
9	4791.702	4785.6816	22.742414	297.6895485	COPDEM30	1006	South Asia East
10	4728.3877	4729.499	33.466225	355.0609837	COPDEM30	293	South Asia East
11	4707.585	4709.9424	24.101397	345.9422789	COPDEM30	322	South Asia East
12	4734.9287	4726.8906	22.013369	333.3874073	COPDEM30	605	South Asia East
13	4964.848	4966.7285	20.479298	222.764267	COPDEM30	825	South Asia East
14	5005.836	5007.835	26.883448	242.300766	COPDEM30	795	South Asia East
15	5011.247	5019.753	25.690332	257.5908661	COPDEM30	506	South Asia East

Only variables that are necessary for glacier inventory analysis were kept. To clean up the data and improve its ease of analysis, unnecessary columns were removed.

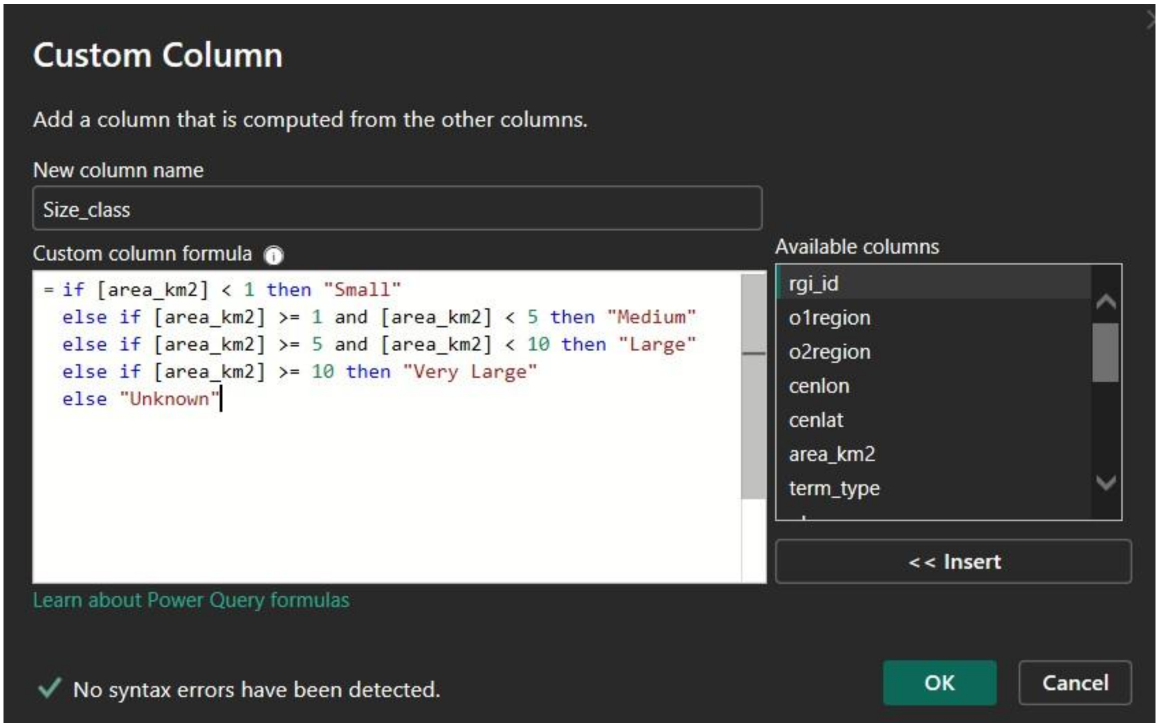
- **Appending three RGI datasets**



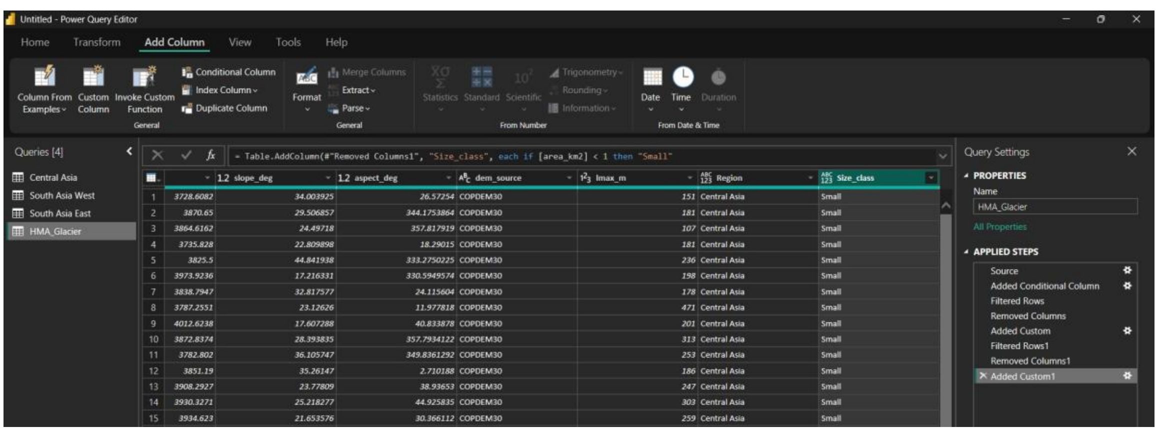
- **Combined HMA_Glacier Dataset**



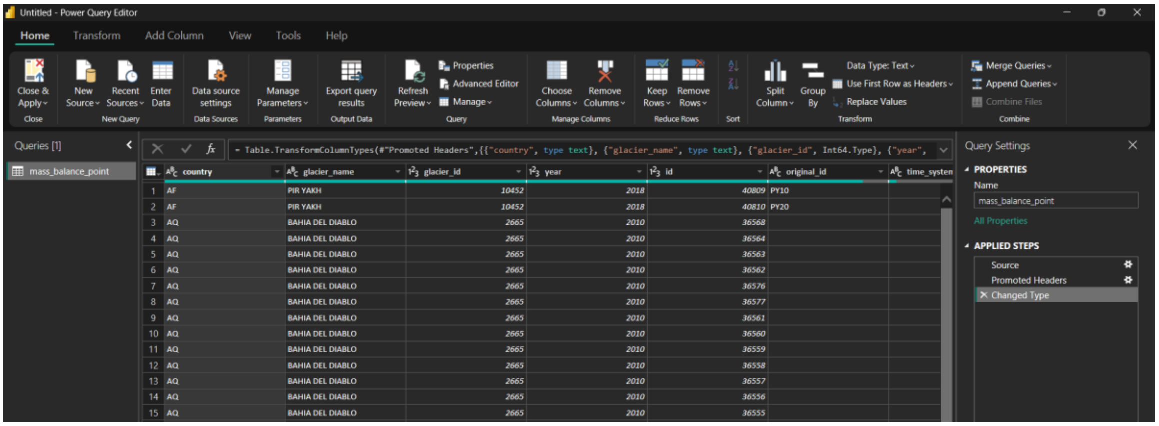
- ## Creating the Size_class conditional column



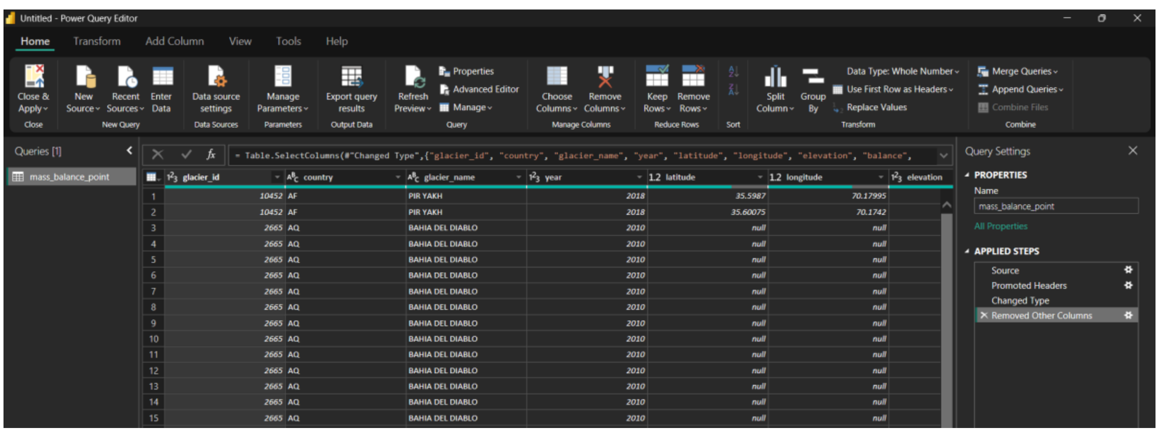
- ## Cleaned Dataset



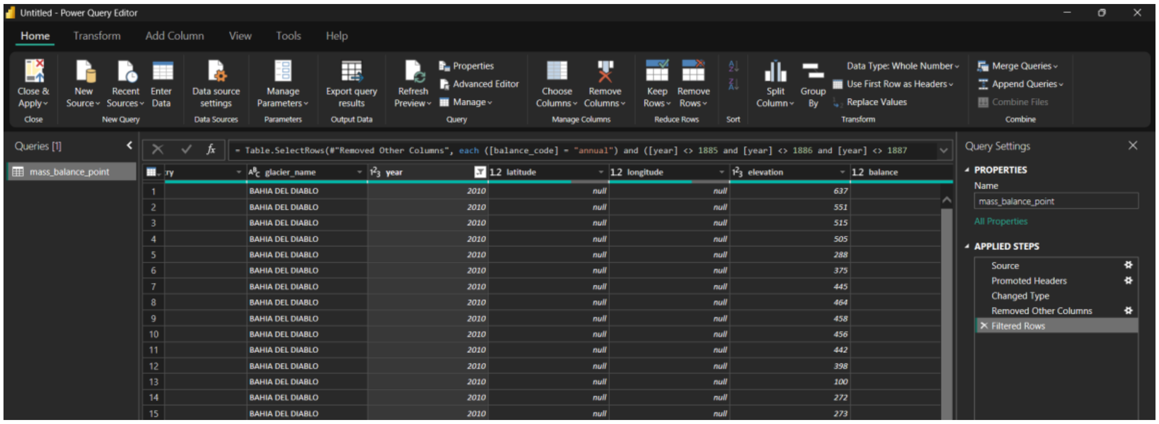
For, Glacier mass-balance dataset



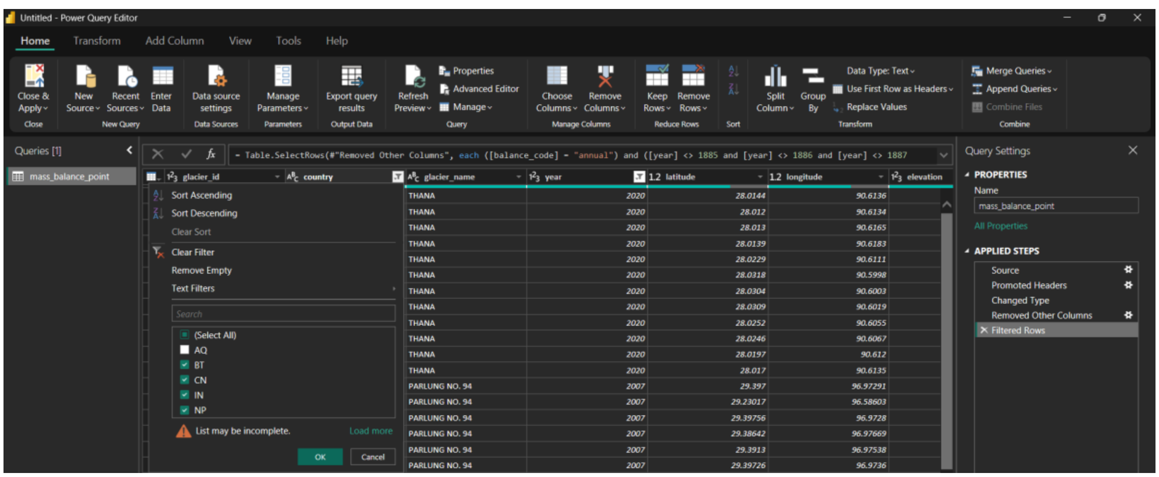
First, we import the mass_balance_point dataset in Power BI. And open the power query editor



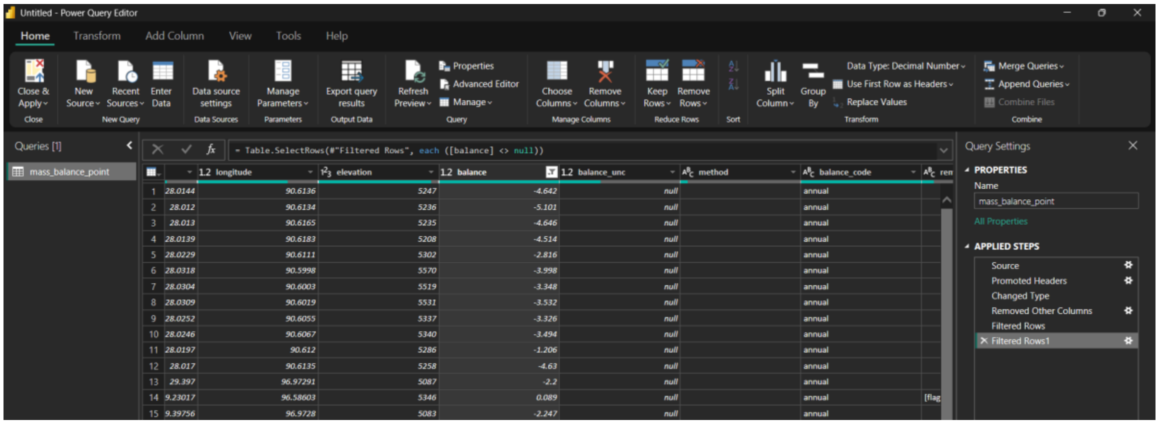
The mass-balance dataset was visually screened for missing and duplicate balances, incorrect data types ,missing balance values and remove unnecessary columns,



The data has been restricted to the study period (1990–2025).

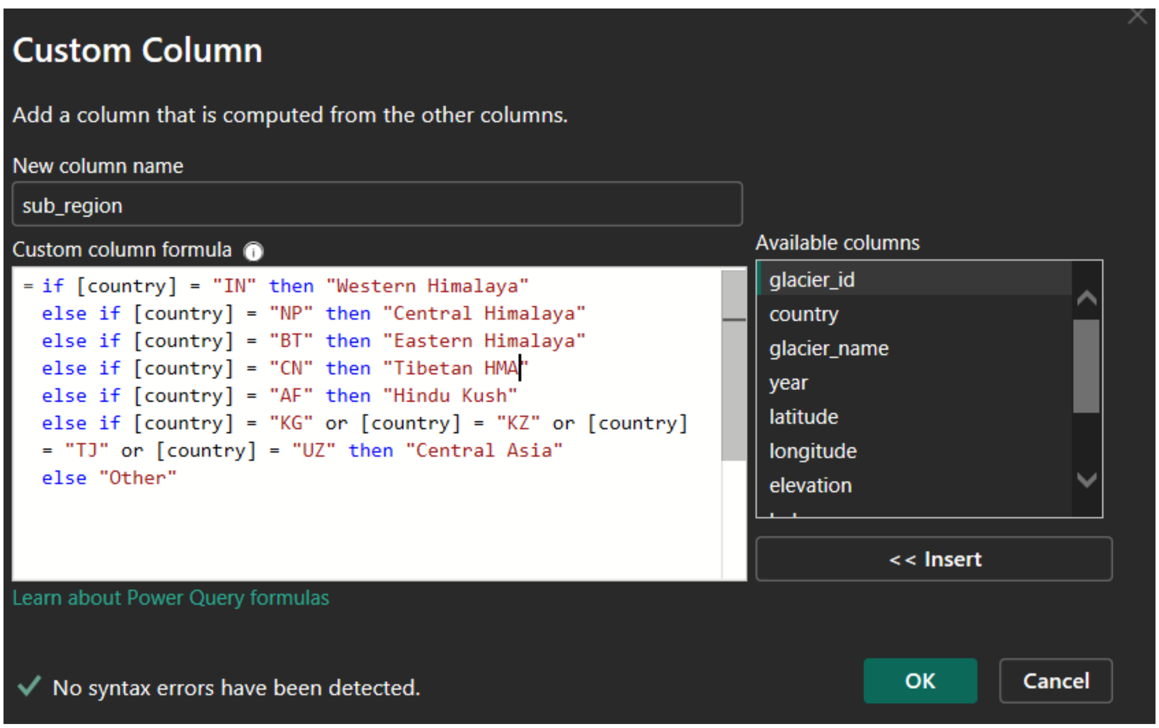


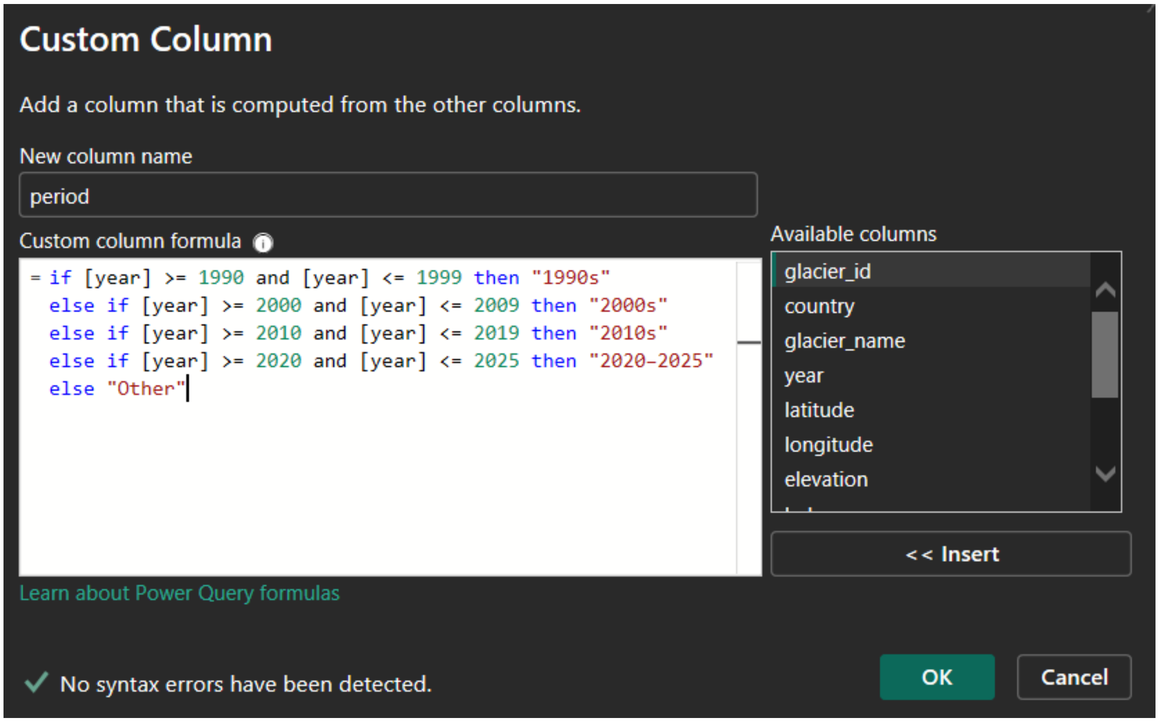
For Himalayan focused region, we only use IN, NP, BT and CN country codes and filtered only annual balance records



Since balance is main variable, remove null values from it.

- ### **Creating sub_region , period variables**





- **Cleaned mass_balance_point dataset**